KUZLAB - KUZAI.ORG

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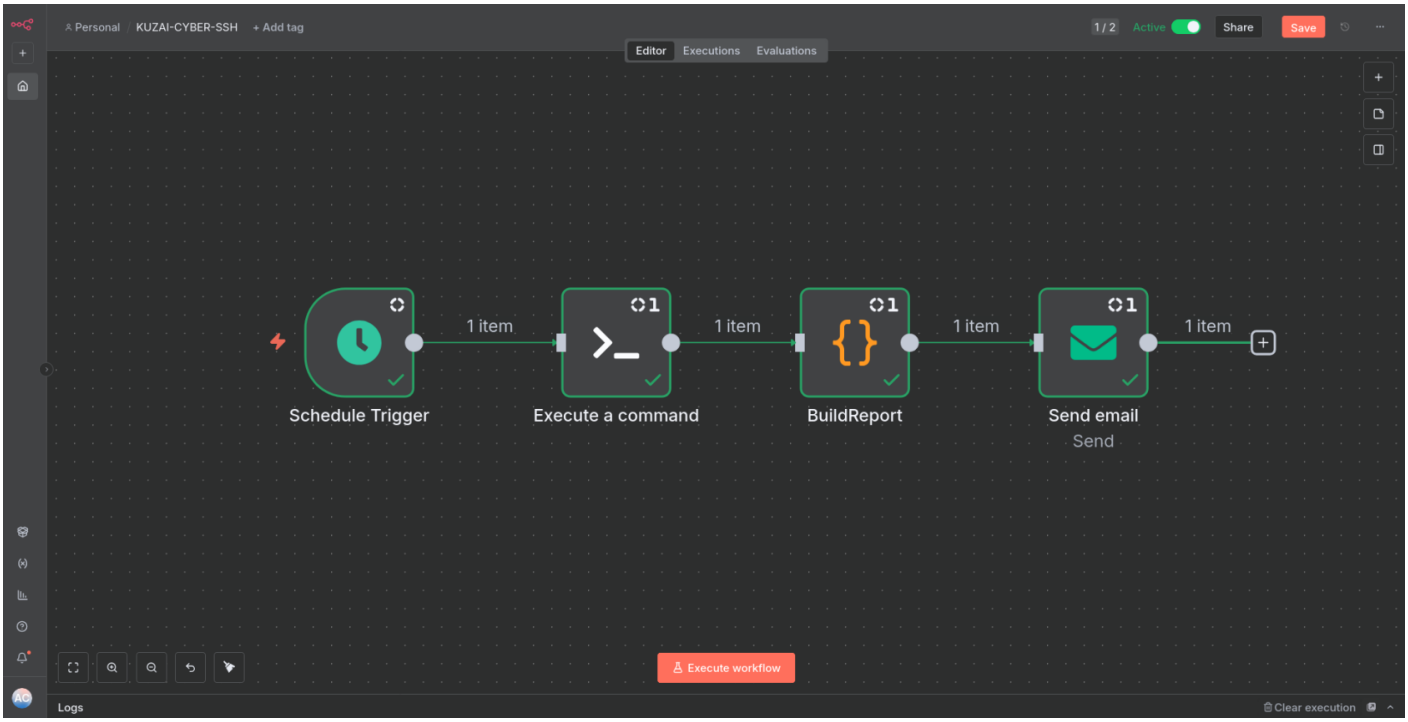
KUZAI.ORG - CYBER - SSH - INTRUSION DETECTION AUTOMATION WORKFLOW

Overview

Developed by the **CYBER TEAM of KUZ NETWORK**, the `KUZAI-CYBER-SSH` project automates the detection and reporting of suspicious SSH activities (both successful and failed login attempts) using `n8n`. It combines Linux system log inspection and email reporting in a robust, modular workflow.



Purpose

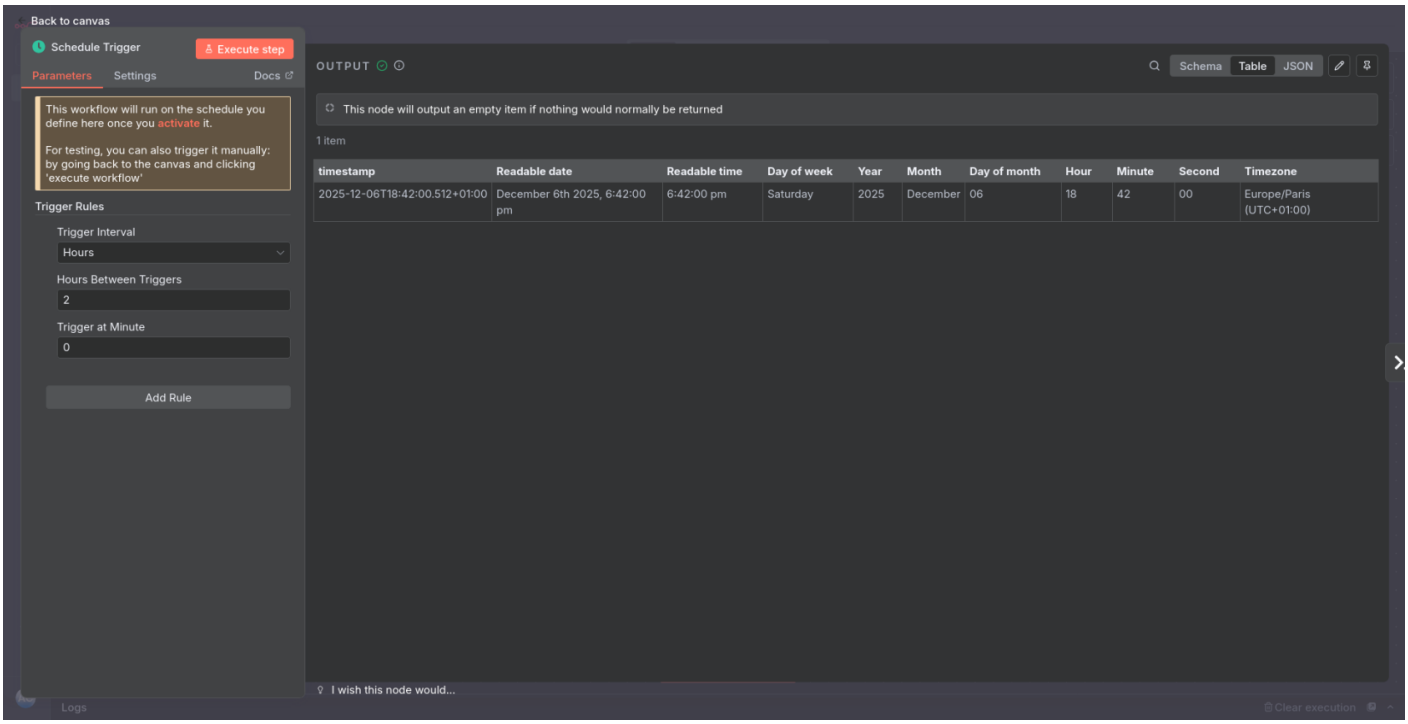
The purpose of this automation is:

- Monitor and summarize SSH access activity over the last 4 hours.
- Detect and display IPs involved in successful and failed login attempts.
- Display total number of banned IPs from `fail2ban`.
- Generate and send a real-time intrusion report via email.

Step-by-Step Installation and Node Details

Node 1: Schedule Trigger

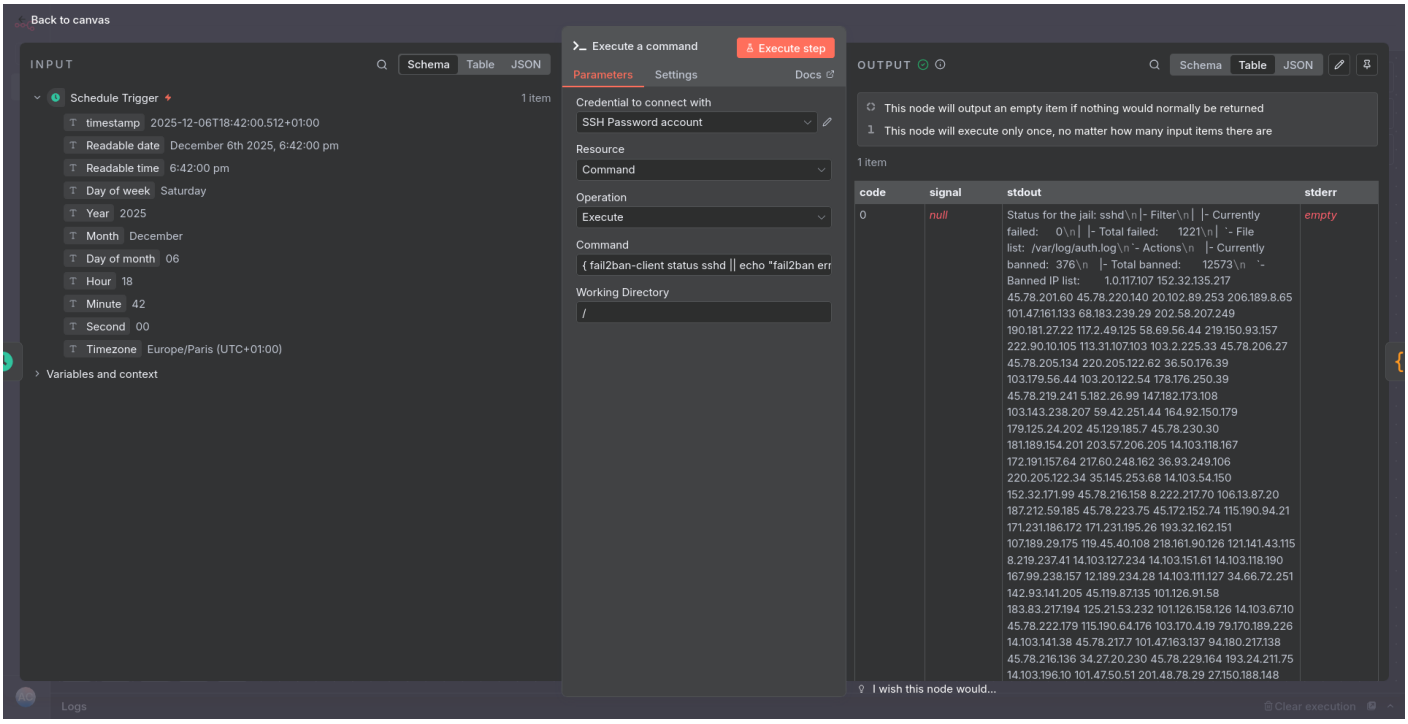
Function: Executes the workflow periodically every 2 hours.



- **Trigger Interval**: Every 2 hours
- **Timezone**: Europe/Paris

Node 2: Execute a command (SSH)

Function: Connects via SSH to a remote Linux system, retrieves key logs and `fail2ban` status.



Shell Command

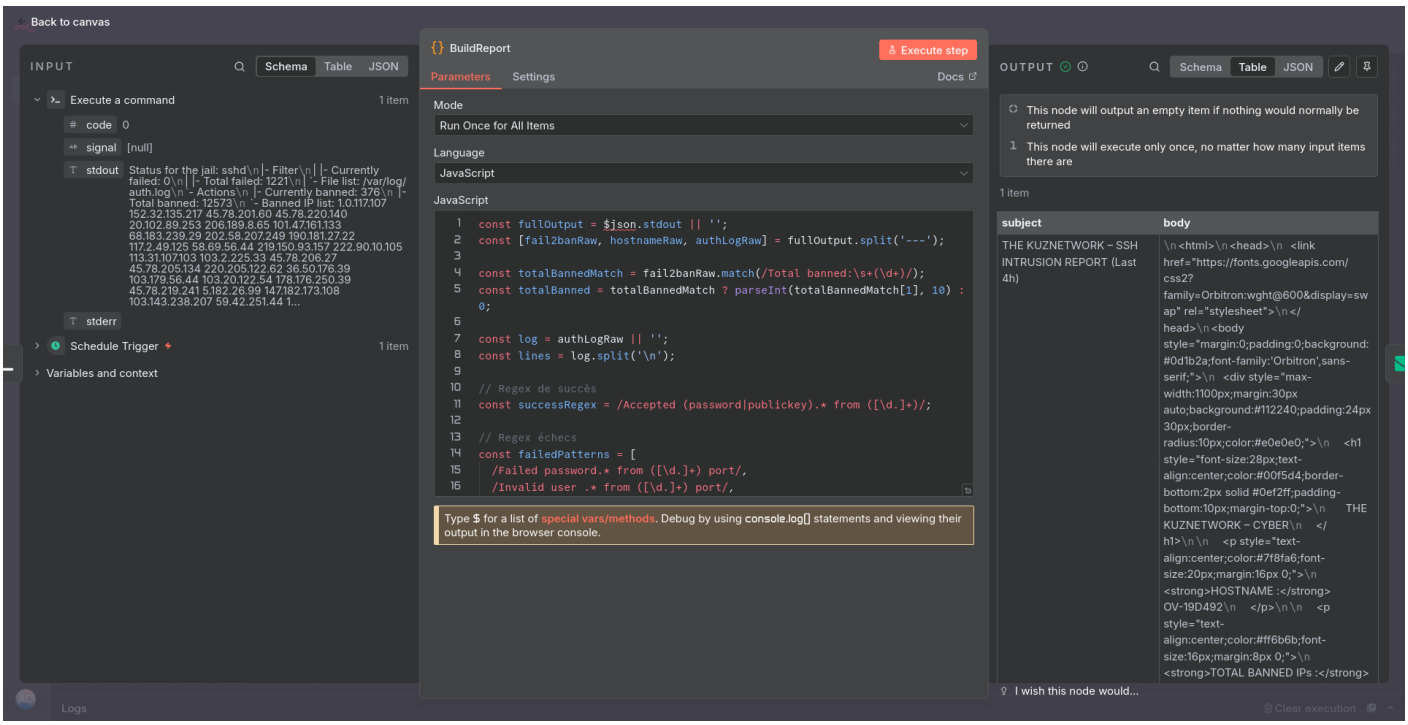
```
{ fail2ban-client status sshd || echo "fail2ban error"; echo '---'; hostname; echo '---'; tail -n 1000 /var/log/auth.log; }
```

Output:

- `fail2ban-client status sshd`: retrieves total number of banned IPs.
- `hostname`: gets the server's hostname.
- `tail -n 1000 /var/log/auth.log`: extracts the last 1000 lines from the SSH log.

Node 3: Build Report (JavaScript)

Function: Parses the raw shell output, extracts and processes login data.



Script Breakdown

```
const fullOutput = $json.stdout || '';
const [fail2banRaw, hostnameRaw, authLogRaw] = fullOutput.split('---');

const totalBannedMatch = fail2banRaw.match(/Total banned:\s+(\d+)/);
const totalBanned = totalBannedMatch ? parseInt(totalBannedMatch[1], 10) : 0;

const log = authLogRaw || '';
const lines = log.split('\n');
```

```
// Regex de succès
const successRegex = /Accepted (password|publickey).* from ([\d.]+)/;

// Regex échecs
const failedPatterns = [
  /Failed password.* from ([\d.]+) port/,
  /Invalid user .* from ([\d.]+) port/,
  /Connection closed by invalid user .* ([\d.]+) port/,
  /Disconnected from authenticating user .* ([\d.]+) port/,
  /Connection reset by ([\d.]+) \[/,
  /Received disconnect from ([\d.]+) port/,
];

// 4 dernières heures
const now = new Date();
const fourHoursAgo = new Date(now.getTime() - 4 * 60 * 60 * 1000);

const success = [];
const failed = [];

for (const line of lines) {
  const timestampMatch = line.match(/^(\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2})/);
  if (!timestampMatch) continue;

  const logTime = new Date(timestampMatch[1]);
  if (isNaN(logTime.getTime()) || logTime < fourHoursAgo) continue;

  const s = line.match(successRegex);
  if (s) {
    const ip = s[2];
    const time = logTime.toISOString().replace('T', ' ').slice(0, 19);
    success.push({ ip, time });
    continue;
  }

  for (const pattern of failedPatterns) {
    const f = line.match(pattern);
    if (f) {
      const ip = f[1];
      const time = logTime.toISOString().replace('T', ' ').slice(0, 19);
```

```
      failed.push({ ip, time });
      break;
    }
  }
}

// Connexions réussies : rendu 2 colonnes
const renderSuccess = () => {
  if (success.length === 0) {
    return `<p style="color:#00f5d4;margin:0 0 12px;font-size:16px;">NONE</p>`;
  }
  const rows = [];
  for (let i = 0; i < success.length; i += 2) {
    const left = `<strong>${success[i].time}</strong> — FROM IP :
<strong>${success[i].ip}</strong>`;
    const right = success[i + 1]
      ? `<strong>${success[i + 1].time}</strong> — FROM IP : <strong>${success[i +
1].ip}</strong>`
      : '';
    rows.push(`<tr><td style="padding:4px 12px;">${left}</td><td style="padding:4px
12px;">${right}</td></tr>`);
  }
  return `
    <p style="color:#00f5d4;margin:0 0 12px;font-size:16px;">
      TOTAL : <strong>${success.length}</strong> ATTEMPT(S)
    </p>
    <table style="width:100%;color:#00f5d4;font-size:16px;margin-top:8px;">
      ${rows.join('')}
    </table>
  `;
};

// Connexions échouées : rendu 2 colonnes
const renderFailed = () => {
  if (failed.length === 0) {
    return `<p style="color:#ff6b6b;margin:0 0 12px;font-size:16px;">NONE</p>`;
  }
  const rows = [];
  for (let i = 0; i < failed.length; i += 2) {
    const left = `<strong>${failed[i].time}</strong> — FROM IP :
```

```
<strong>${failed[i].ip}</strong>`;
    const right = failed[i + 1]
      ? `<strong>${failed[i + 1].time}</strong> — FROM IP : <strong>${failed[i +
1].ip}</strong>`
      : '';
    rows.push(`<tr><td style="padding:4px 12px;">${left}</td><td style="padding:4px
12px;">${right}</td></tr>`);
  }
  return `
    <p style="color:#ff6b6b;margin:0 0 12px;font-size:16px;">
      TOTAL : <strong>${failed.length}</strong> ATTEMPT(S)
    </p>
    <table style="width:100%;color:#ff6b6b;font-size:16px;margin-top:8px;">
      ${rows.join('')}
    </table>
  `;
};

// Format date
const formattedDate = now.toLocaleString('fr-FR', {
  year: 'numeric',
  month: 'long',
  day: 'numeric',
  hour: '2-digit',
  minute: '2-digit',
  second: '2-digit',
  timeZone: 'Europe/Paris',
  hour12: false
}).toUpperCase();

const hostname = (hostnameRaw || '').trim().toUpperCase();

const html = `
<html>
<head>
  <link href="https://fonts.googleapis.com/css2?family=Orbitron:wght@600&display=swap"
rel="stylesheet">
</head>
<body style="margin:0;padding:0;background:#0d1b2a;font-family:'Orbitron',sans-serif;">
  <div style="max-width:1100px;margin:30px auto;background:#112240;padding:24px 30px;border-
```

```
radius:10px;color:#e0e0e0;">
    <h1 style="font-size:28px;text-align:center;color:#00f5d4;border-bottom:2px solid
#0ef2ff;padding-bottom:10px;margin-top:0;">
      THE KUZNETWORK — CYBER
    </h1>

    <p style="text-align:center;color:#7f8fa6;font-size:20px;margin:16px 0;">
      <strong>HOSTNAME :</strong> ${hostname}
    </p>

    <p style="text-align:center;color:#ff6b6b;font-size:16px;margin:8px 0;">
      <strong>TOTAL BANNED IPs :</strong> ${totalBanned}
    </p>

    <h3 style="color:#00f5d4;margin-top:32px;font-size:20px;">□ SUCCESSFUL SSH LOGINS</h3>
    ${renderSuccess()}

    <h3 style="color:#ff6b6b;margin-top:32px;font-size:20px;">□ FAILED SSH LOGIN ATTEMPTS</h3>
    ${renderFailed()}

    <hr style="margin-top:40px;border:0;border-top:1px solid #2b3e5e;">
    <p style="text-align:center;font-size:13px;color:#7f8fa6;margin-top:16px;">
      REPORT GENERATED ON <strong>${formattedDate}</strong>
    </p>
  </div>
</body>
</html>
`;

return [{
  json: {
    subject: 'THE KUZNETWORK — SSH INTRUSION REPORT (Last 4h)',
    body: html
  }
}];
```

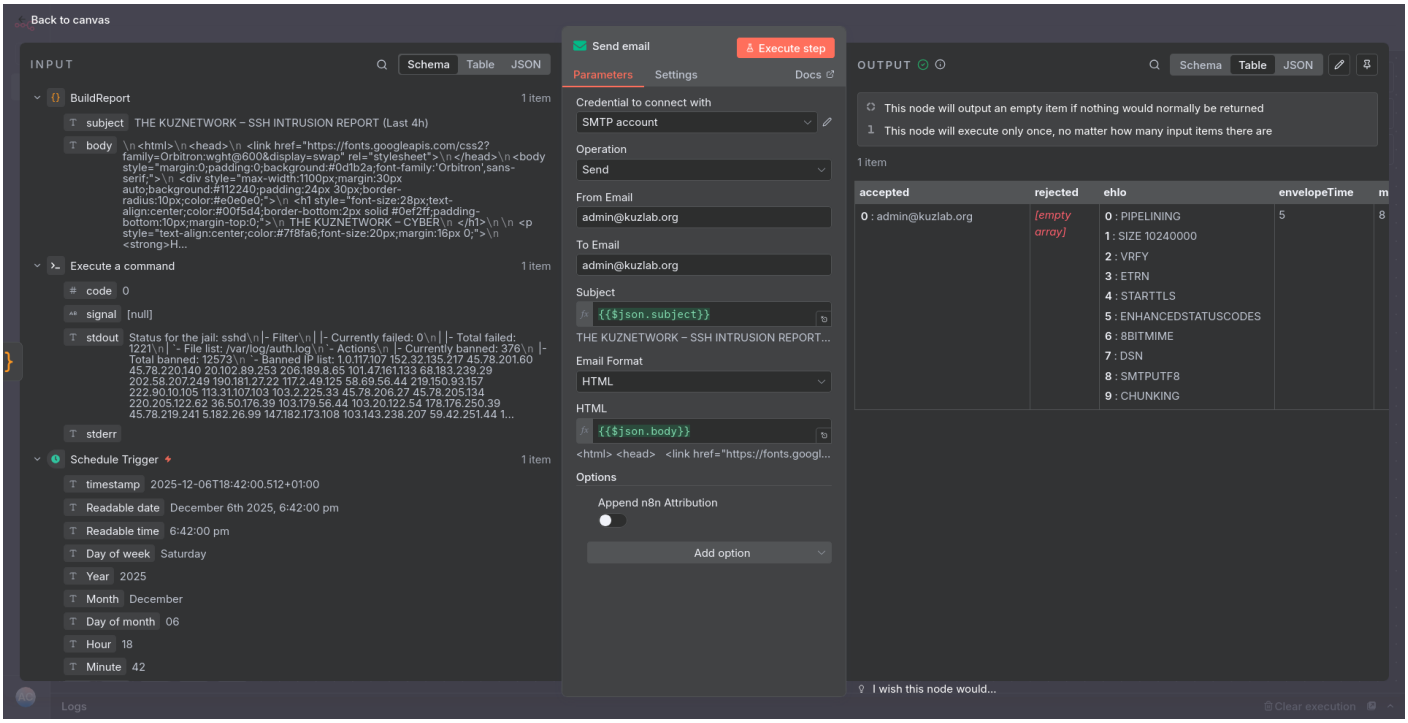
Output

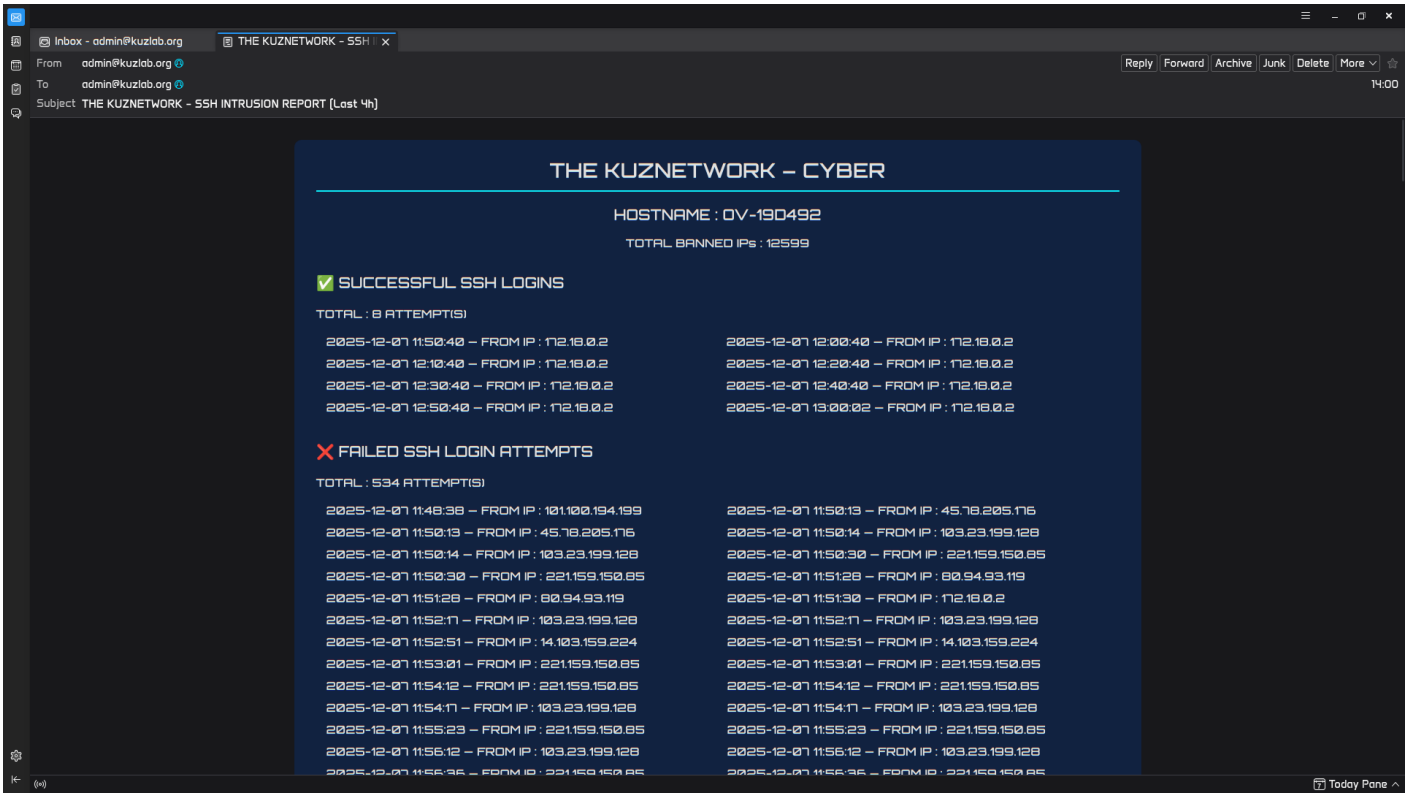
- Groups successful attempts by IP address and counts them.
- Lists each failed attempt with timestamp and IP.
- Extracts total number of banned IPs.

- Stores hostname and date of report.

Node 4: Send Email

Function: Sends the final HTML report to an administrator inbox.





- **SMTP Email**: admin@kuzlab.org
- **Format**: HTML
- **Attribution**: n8n footer disabled

Email Report Output

Structure:

HTML Output

- Title: KUZNETWORK – CYBER
- Hostname and Banned IP count
- 🟢 Successful SSH Logins (in 2-column layout)
- 🔴 Failed SSH Login Attempts (in 2-column layout)
- Timestamp of generation

Deployment Checklist

1. Install and configure `n8n` with SSH and SMTP credentials.
2. Add the four nodes:
   - Schedule Trigger (every 2h)
   - Execute Command (custom Linux shell command)
   - Build Report (JavaScript processor)
   - Send Email (HTML format, no attribution)
3. Enable the workflow.

Additional Notes

- The system is timezone aware (`Europe/Paris`).
- SSH connection uses credentials defined in `Execute Command`.
- Make sure the host allows `fail2ban-client` and SSH log reading.
- Font: Orbitron (Google Fonts) is used for a cyber-style look.
- Works with `auth.log` (Debian/Ubuntu). Adapt for `secure` on RHEL/CentOS.

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KUZAI.ORG - CHATBOT - FULL INSTALL PROCESS

Cloudflare Workers AI + WordPress

0) Project overview

What you are deploying

You will add a floating chatbot widget on `kuzai.org` (WordPress).
The widget sends messages to a Cloudflare Worker API endpoint.
The Worker calls Cloudflare Workers AI (LLM inference on Cloudflare) and returns a reply to the widget.

Why this architecture

- The website server only serves WordPress + plugin files.
- LLM inference runs on Cloudflare (not on your server).
- No API key is stored inside WordPress.
- This works on Cloudflare Free (within quotas).

Cloudflare Free limits (important)

Two distinct quotas can stop replies:

- **Workers execution requests**: ~100,000 requests/day (resets at 00:00 UTC)
- **Workers AI "neurons/day"**: ~10,000 neurons/day (then typically HTTP 429 until reset or upgrade)

When limits are hit:

- The Worker may return **HTTP 429**
- The widget will show an error instead of a reply

Security model

- No secret key in WordPress.
- The Worker restricts CORS to:
  - `https://kuzai.org`
  - `https://www.kuzai.org`
- This process stores nothing by design (no DB logging).

---

1) Cloudflare — Create a Worker + API endpoint

1.1 Create the Worker

Do this:

1. Open Cloudflare Dashboard
2. Go to **Workers & Pages**
3. Click **Create Worker**
4. Name: `kuzai-chat`
5. Click **Deploy**

1.2 Add Workers AI binding (mandatory)

Topo: this is what gives your Worker access to `env.AI.run()`.

Do this:

1. Worker → **Settings** → **Bindings** → **Add binding**
2. Type: **AI**
3. Name: **AI**
4. Save, then **redeploy**

1.3 Worker code (FULL)

Topo: this exposes a single API route `/api/chat`, validates input, enforces CORS, then runs an LLM.

Do this:

1. Open the Worker editor
2. Replace everything with the code below

3. Save + deploy

```
export default {
  async fetch(request, env) {
    const url = new URL(request.url);

    // CORS: allow only your site
    const allowedOrigins = new Set([
      "https://kuzai.org",
      "https://www.kuzai.org",
    ]);

    const origin = request.headers.get("Origin") || "";
    const corsHeaders = allowedOrigins.has(origin)
      ? {
          "Access-Control-Allow-Origin": origin,
          "Access-Control-Allow-Methods": "POST, OPTIONS",
          "Access-Control-Allow-Headers": "Content-Type",
          "Vary": "Origin",
        }
      : {};

    // Preflight (browser sends this automatically)
    if (request.method === "OPTIONS") {
      return new Response(null, { status: 204, headers: corsHeaders });
    }

    // Only accept POST /api/chat
    if (url.pathname !== "/api/chat" || request.method !== "POST") {
      return new Response("Not Found", { status: 404, headers: corsHeaders });
    }

    // Parse JSON body
    let data;
    try {
      data = await request.json();
    } catch {
      return new Response(JSON.stringify({ error: "invalid_json" }), {
        status: 400,
        headers: { "Content-Type": "application/json", ...corsHeaders },
```

```
      });
    }

    // Validate message
    const message = String(data?.message || "").trim();
    if (!message) {
      return new Response(JSON.stringify({ error: "empty_message" }), {
        status: 400,
        headers: { "Content-Type": "application/json", ...corsHeaders },
      });
    }
    if (message.length > 2000) {
      return new Response(JSON.stringify({ error: "message_too_long" }), {
        status: 400,
        headers: { "Content-Type": "application/json", ...corsHeaders },
      });
    }

    // KUZAI-scoped system prompt
    const systemPrompt = `You are the KUZAI Assistant for kuzai.org.
Your job is to help visitors navigate AI tools and categories.
Do NOT invent tools, URLs, or data.
Be concise, technical, and neutral.
If uncertain, say you don't know and suggest using site navigation/categories/search.
Do not ask for secrets or personal data.`;

    try {
      const result = await env.AI.run("@cf/meta/llama-3.1-8b-instruct", {
        messages: [
          { role: "system", content: systemPrompt },
          { role: "user", content: message },
        ],
        temperature: 0.3,
        max_tokens: 350,
      });

      return new Response(JSON.stringify({ reply: result?.response || "" }), {
        status: 200,
        headers: { "Content-Type": "application/json", ...corsHeaders },
      });
```

```
    } catch (e) {
      // Quota exceeded or model unavailable often ends here
      return new Response(JSON.stringify({ error: "ai_unavailable" }), {
        status: 502,
        headers: { "Content-Type": "application/json", ...corsHeaders },
      });
    }
  },
};
```

1.4 Endpoint to use

Topo: your WordPress plugin will call this URL.

- Worker base example:
  - `https://kuzai-chat.elijahkaminski230.workers.dev`
- API endpoint:
  - `https://kuzai-chat.elijahkaminski230.workers.dev/api/chat`

---

2) Server — WordPress plugin install

2.1 Create plugin directories

Topo: this creates a standard WordPress plugin layout: one PHP file + assets.

Do this:

```
mkdir -p /var/www/html/kuzai.org/wp-content/plugins/kuzai-chatbot/assets
```

---

3) Plugin code — PHP + JS + CSS (FULL)

3.1 PHP plugin file (FULL)

Topo: this registers CSS/JS, injects the API URL into JS, and provides a shortcode that prints the widget HTML.

Do this:

1. Create/edit:
   - `/var/www/html/kuzai.org/wp-content/plugins/kuzai-chatbot/kuzai-chatbot.php`
2. Paste everything below:

```
<?php
/*
Plugin Name: KUZAI Chatbot (Workers AI)
Description: Floating chatbot widget calling Cloudflare Worker API.
Version: 1.2.0
*/

if (!defined('ABSPATH')) exit;

/**
 * Load CSS/JS assets on the frontend.
 * - CSS for the neon UI
 * - JS for open/close, expand, fetch API calls
 */
function kuzai_chatbot_assets() {
  $plugin_url = plugin_dir_url(__FILE__);

  // Load CSS
  wp_enqueue_style(
    'kuzai-chatbot-css',
    $plugin_url . 'assets/kuzai-chatbot.css',
    array(),
    '1.2.0'
  );

  // Load JS (footer)
  wp_enqueue_script(
    'kuzai-chatbot-js',
    $plugin_url . 'assets/kuzai-chatbot.js',
    array(),
    '1.2.0',
    true
```

```
  );

  // Provide API endpoint to JS (no secrets here)
  wp_localize_script('kuzai-chatbot-js', 'KUZAI_CHATBOT', array(
    'api' => 'https://kuzai-chat.elijahkaminski230.workers.dev/api/chat'
  ));
}
add_action('wp_enqueue_scripts', 'kuzai_chatbot_assets');

/**
 * Shortcode output:
 * - Floating button
 * - Chat window with header (expand + close)
 * - Log area
 * - Input form
 */
function kuzai_chatbot_shortcode() {
  ob_start(); ?>

  <button type="button" id="kuzaiChatBtn">KUZCHAT</button>

  <div id="kuzaiChatBox" role="dialog" aria-label="KUZAI Assistant">
    <div id="kuzaiChatHdr">
      <span>KUZAI ASSISTANT</span>
      <div>
        <button type="button" id="kuzaiChatExpand" title="Expand / Reduce">□</button>
        <button type="button" id="kuzaiChatClose" title="Close">×</button>
      </div>
    </div>

    <div id="kuzaiChatLog"></div>

    <form id="kuzaiChatForm">
      <input id="kuzaiChatInput" placeholder="Type your request..." autocomplete="off" />
      <button id="kuzaiChatSend" type="submit">SEND</button>
    </form>
  </div>

  <?php
  return ob_get_clean();
```

```
}
add_shortcode('kuzai_chatbot', 'kuzai_chatbot_shortcode');
```

Update the API URL (mandatory)

Topo: the plugin must point to your Worker.

Edit this line inside the PHP file:

```
'api' => 'https://kuzai-chat.elijahkaminski230.workers.dev/api/chat'
```

Replace with your Worker URL if different.

3.2 JavaScript file (FULL)

Topo: this handles open/close, expand, ESC-to-close, message display, and fetch() calls.

Do this:

1. Create/edit:
   - `/var/www/html/kuzai.org/wp-content/plugins/kuzai-chatbot/assets/kuzai-chatbot.js`
2. Paste:

```
(function () {
  function init() {
    const btn = document.getElementById("kuzaiChatBtn");
    const box = document.getElementById("kuzaiChatBox");
    const closeBtn = document.getElementById("kuzaiChatClose");
    const expandBtn = document.getElementById("kuzaiChatExpand");
    const log = document.getElementById("kuzaiChatLog");
    const form = document.getElementById("kuzaiChatForm");
    const input = document.getElementById("kuzaiChatInput");

    // If shortcode not present on this page, do nothing
    if (!btn || !box || !log || !form || !input) return;

    // Prevent double init if shortcode is accidentally duplicated
    if (window.__kuzaiChatInitDone) return;
    window.__kuzaiChatInitDone = true;

    const API_URL = (window.KUZAI_CHATBOT && window.KUZAI_CHATBOT.api) || "";
```

```
    if (!API_URL) return;

    // Initial state: closed
    box.style.display = "none";
    box.classList.remove("kuzai-expanded");
    if (expandBtn) expandBtn.textContent = "□";

    function addMsg(cls, prefix, txt) {
      const p = document.createElement("p");
      p.className = "kmsg " + cls;
      p.textContent = prefix + txt;
      log.appendChild(p);
      log.scrollTop = log.scrollHeight;
    }

    function ensureWelcome() {
      if (log.childNodes.length === 0) {
        addMsg(
          "kai",
          "KUZAI --> ",
          "Hello. I'm the KUZAI Assistant. Describe your needs, and I will guide you toward
relevant KUZAI tools and categories."
        );
      }
    }

    function openBox() {
      // Must be flex to keep layout (CSS uses flex column)
      box.style.display = "flex";
      ensureWelcome();
      input.focus();
    }

    function closeBox() {
      box.classList.remove("kuzai-expanded");
      if (expandBtn) expandBtn.textContent = "□";
      box.style.display = "none";
    }

    function isOpen() {
```

```
  return box.style.display === "flex" || box.style.display === "block";
}

// Toggle open/close
btn.addEventListener("click", function () {
  isOpen() ? closeBox() : openBox();
});

// Close button
if (closeBtn) closeBtn.addEventListener("click", closeBox);

// Expand/Reduce button (80% viewport via CSS class)
if (expandBtn) {
  expandBtn.addEventListener("click", function () {
    const expanded = !box.classList.contains("kuzai-expanded");
    box.classList.toggle("kuzai-expanded", expanded);
    expandBtn.textContent = expanded ? "□" : "□";
  });
}

// ESC closes
document.addEventListener("keydown", function (e) {
  if (e.key === "Escape") closeBox();
});

// Send message
form.addEventListener("submit", async function (e) {
  e.preventDefault();
  const msg = (input.value || "").trim();
  if (!msg) return;

  input.value = "";
  addMsg("kme", "ME --> ", msg);

  // Placeholder while waiting
  addMsg("kai", "KUZAI --> ", "…");

  try {
    const r = await fetch(API_URL, {
      method: "POST",
```

```
          headers: { "Content-Type": "application/json" },
          body: JSON.stringify({ message: msg }),
        });

        const data = await r.json().catch(() => ({}));

        // Remove placeholder
        if (log.lastChild) log.removeChild(log.lastChild);

        if (!r.ok) {
          addMsg("kai", "KUZAI --> ", "[error] " + (data.error || ("HTTP " + r.status)));
          return;
        }

        addMsg("kai", "KUZAI --> ", (data.reply || "[no reply]"));
      } catch (err) {
        if (log.lastChild) log.removeChild(log.lastChild);
        addMsg("kai", "KUZAI --> ", "[backend error]");
      }
    });
  }

  // Init when DOM is ready
  if (document.readyState === "loading") {
    document.addEventListener("DOMContentLoaded", init);
  } else {
    init();
  }
})();
```

3.3 CSS file (FULL)

Topo: this is the neon UI theme, with stable flex layout and an “expand” mode.

Do this:

1. Create/edit:
   - `/var/www/html/kuzai.org/wp-content/plugins/kuzai-chatbot/assets/kuzai-chatbot.css`
2. Paste:

```
/* ========================================================
   KUZAI CHATBOT — NEON THEME (STABLE LAYOUT)
   - Flex column window
   - Log uses remaining space (flex: 1)
   - Input pinned to bottom
   - Expand mode ~80% viewport
   ======================================================== */

/* Floating button */
#kuzaiChatBtn{
  position:fixed;
  right:18px;
  bottom:18px;
  z-index:99999;

  padding:10px 12px;
  border-radius:12px;

  border:1px solid rgba(0,204,255,.45);
  background:rgba(0,10,20,.78);
  color:#d8f4ff;

  font-weight:900;
  letter-spacing:.5px;
  cursor:pointer;

  backdrop-filter:blur(10px);
  box-shadow:0 0 18px rgba(0,204,255,.18);
}

/* Chat window (closed by default; JS sets display:flex) */
#kuzaiChatBox{
  position:fixed;
  right:18px;
  bottom:70px;

  width:450px;
  max-width:92vw;
  height:650px;
  max-height:80vh;
```

```
  z-index:99999;

  border-radius:18px;
  border:1px solid rgba(0,204,255,.38);
  background:rgba(0,8,18,.88);

  backdrop-filter:blur(12px);
  box-shadow:0 0 28px rgba(0,204,255,.14);

  overflow:hidden;

  display:none;
  flex-direction:column;
}

/* Header */
#kuzaiChatHdr{
  padding:12px 12px;
  border-bottom:1px solid rgba(0,204,255,.18);
  color:#e9fbff;

  font-weight:900;
  letter-spacing:.5px;

  display:flex;
  align-items:center;
  justify-content:space-between;

  background:linear-gradient(180deg, rgba(0,204,255,.10), rgba(0,0,0,0));
}

/* Buttons container: force right alignment */
#kuzaiChatHdr > div{
  margin-left:auto;
  display:flex;
  align-items:center;
  gap:6px;
}
```

```
/* Header button size control */
#kuzaiChatHdr button{
  transition: background .2s ease, transform .15s ease;

  width:35px;
  height:35px;
  padding:0;

  display:flex;
  align-items:center;
  justify-content:center;

  border-radius:8px;
  cursor:pointer;

  border:1px solid rgba(0,204,255,.25);
  background:rgba(0,0,0,.25);
  color:#cfefff;

  font-size:18px;
}

#kuzaiChatHdr button:hover{
  background:rgba(0,204,255,.12);
  transform:scale(1.08);
}

#kuzaiChatExpand{ font-size:18px; }
#kuzaiChatClose{ font-size:18px; font-weight:900; }

/* Log area takes remaining height */
#kuzaiChatLog{
  padding:12px 12px;

  flex:1;
  min-height:0;
  overflow:auto;

  color:#d8f4ff;
  font-size:13px;
```

```
  line-height:1.45;

  background:
    radial-gradient(800px 300px at 20% 0%, rgba(0,204,255,.06), transparent 55%),
    radial-gradient(600px 240px at 80% 20%, rgba(0,120,255,.06), transparent 60%);
}

/* Bottom form pinned */
#kuzaiChatForm{
  display:flex;
  gap:10px;

  padding:12px 12px;
  border-top:1px solid rgba(0,204,255,.18);

  margin-top:auto;

  background:rgba(0,0,0,.18);
}

/* Input */
#kuzaiChatInput{
  flex:1;

  border-radius:12px;
  border:1px solid rgba(0,204,255,.22);
  background:rgba(0,0,0,.35);

  color:#e9fbff;
  padding:10px 12px;
  outline:none;
}

#kuzaiChatInput:focus{
  border-color:rgba(0,204,255,.55);
  box-shadow:0 0 0 3px rgba(0,204,255,.12);
}

/* Send button */
#kuzaiChatSend{
```

```
  border-radius:12px;
  border:1px solid rgba(0,204,255,.38);
  background:rgba(0,204,255,.10);

  color:#e9fbff;
  font-weight:900;

  padding:10px 14px;
  cursor:pointer;
}

#kuzaiChatSend:hover{
  background:rgba(0,204,255,.16);
  box-shadow:0 0 16px rgba(0,204,255,.18);
}

/* Messages */
.kmsg{ margin:0 0 10px; white-space:pre-wrap; }
.kme{ color:#ffffff; }
.kai{ color:#00ccff; }

/* Expand mode (80% viewport) */
#kuzaiChatBox.kuzai-expanded{
  width:80vw !important;
  height:80vh !important;
  max-width:80vw !important;
  max-height:80vh !important;

  right:10vw !important;
  bottom:10vh !important;
}

/* Mobile adjustments */
@media (max-width:480px){
  #kuzaiChatBox{
    right:12px;
    left:12px;
    width:auto;
    height:72vh;
    bottom:64px;
```

```
  }
  #kuzaiChatBtn{
    right:12px;
    bottom:12px;
  }
  #kuzaiChatBox.kuzai-expanded{
    width:96vw !important;
    height:90vh !important;
    right:2vw !important;
    bottom:5vh !important;
  }
}
```

---

4) WordPress — Activate + place shortcode

4.1 Activate plugin

Topo: until activation, WordPress will not load your CSS/JS and the shortcode will do nothing.

Do this:

- WordPress Admin → **Plugins**
- Activate: **KUZAI Chatbot (Workers AI)**

4.2 Add shortcode (place ONCE)

Topo: the widget uses fixed DOM IDs, so duplicating it on one page can break the JS.

Do this:

- In a page, footer block, or global area, add:

```
[kuzai_chatbot]
```

---

5) Test

5.1 Test the Worker API from your server

Topo: this validates the API independently of WordPress.

```
curl -s -X POST "https://kuzai-chat.elijahkaminski230.workers.dev/api/chat"   -H "Content-Type: application/json"   -H "Origin: https://kuzai.org"   -d '{"message":"AI tool to analyze Apache logs?"}'
```

Expected output:

- JSON like: `{"reply":"..."}`

5.2 Frontend test

Do this:

1. Open `https://kuzai.org`
2. Click **KUZCHAT**
3. Validate:
   - welcome message appears
   - replies are returned
   - expand button toggles size
   - close button closes window

---

6) Common troubleshooting

6.1 “Not Found”

- Cause: calling `/` instead of `/api/chat`
- Fix: use the exact endpoint: `/api/chat`

6.2 “HTTP 429”

- Cause: Cloudflare Free quota hit (Workers AI and/or Workers)
- Fix: wait for reset (00:00 UTC) or upgrade; add rate-limiting later if needed

6.3 “ai_unavailable”

Most common causes:

- Workers AI binding missing or wrong name (must be `AI`)
- Workers AI temporary outage
- quota exceeded

KUZAI.ORG - GITHUB SEARCH ENGINE - MODAL WINDOW

GitHub Search Integration for WordPress via n8n (Complete Guide)

Overview

This document describes how to add a **GitHub search field** to a WordPress site that queries the **GitHub API via n8n** and displays results in a **modal window**.
The architecture ensures **no secrets are exposed client-side** and is designed to be reusable and extensible.

Goal

Add a search field on a WordPress site that:

- Sends search queries to n8n
- n8n securely calls the GitHub API
- Results are returned as JSON
- WordPress displays results in a modal window

Prerequisites

- One Debian VPS for **n8n** (e.g. `n8n.your-domain.com`)
- Apache as reverse proxy with Let’s Encrypt SSL
- n8n running in Docker
- One WordPress site (e.g. `your-domain.com`)

- GitHub Personal Access Token (PAT)
- One shared secret between WordPress and n8n

STEP 1 — Install n8n with Docker on Debian

System packages

```
apt update && apt -y install docker.io docker-compose-plugin apache2 certbot python3-certbot-apache apache2-utils jq
```

Directory structure

```
mkdir -p /root/n8n/{data,files}
cd /root/n8n
```

Generate encryption key

```
export N8N_ENCRYPTION_KEY=$(openssl rand -hex 32)
echo "N8N_ENCRYPTION_KEY=$N8N_ENCRYPTION_KEY"
```

Docker Compose configuration

Create `/root/n8n/docker-compose.yml`:

```
version: "3.8"

services:
  postgres:
    image: postgres:16
    environment:
      POSTGRES_USER: n8n
      POSTGRES_PASSWORD: n8npass
      POSTGRES_DB: n8n
    volumes:
```

```
      - ./db:/var/lib/postgresql/data
    restart: unless-stopped

  n8n:
    image: n8nio/n8n:latest
    user: "1000:1000"
    depends_on:
      - postgres
    environment:
      N8N_PUSH_BACKEND: websocket
      N8N_EDITOR_BASE_URL: https://n8n.your-domain.com/
      N8N_RUNNERS_ENABLED: "true"
      N8N_ENCRYPTION_KEY: "${N8N_ENCRYPTION_KEY}"
      N8N_HOST: n8n.your-domain.com
      N8N_PORT: 5678
      N8N_PROTOCOL: https
      WEBHOOK_URL: https://n8n.your-domain.com/
      DB_TYPE: postgresdb
      DB_POSTGRESDB_HOST: postgres
      DB_POSTGRESDB_DATABASE: n8n
      DB_POSTGRESDB_USER: n8n
      DB_POSTGRESDB_PASSWORD: n8npass
      GENERIC_TIMEZONE: Europe/Paris
      N8N_SECURE_COOKIE: "true"
      N8N_DIAGNOSTICS_ENABLED: "false"
    ports:
      - "127.0.0.1:5678:5678"
    volumes:
      - ./data:/home/node/.n8n
    restart: unless-stopped
```

Start services

```
docker compose up -d
docker compose logs -f n8n
```

Health check

```
curl -i http://127.0.0.1:5678/healthz
```

STEP 2 — Apache Reverse Proxy with SSL

Enable modules and SSL

```
a2enmod proxy proxy_http proxy_wstunnel rewrite headers ssl
certbot --apache -d n8n.your-domain.com --redirect
```

Protect n8n editor with Basic Auth

```
htpasswd -c /etc/apache2/.htpasswd-n8n admin
```

VirtualHost configuration

Create `/etc/apache2/sites-available/n8n.your-domain.com.conf`
(unchanged from original, see appendix if needed)

Reload Apache:

```
apache2ctl configtest && systemctl reload apache2
```

STEP 3 — GitHub Personal Access Token

- Create a **fine-grained or classic PAT**
- Public repositories require no special scopes
- In n8n, create **Bearer Auth credentials**
  - Name: `GitHub PAT`
  - Value: token only
  - Allowed domain: `api.github.com`

STEP 4 — n8n Workflow (4 nodes)

Workflow chain

```
Webhook → Code (Prepare) → HTTP Request → Code (Map)
```

Webhook node

- Method: POST
- Path: `kuz/github-search`
- Authentication: None
- Response: When last node finishes

Code node — Prepare

(Original JavaScript preserved)

HTTP Request node

- GET `https://api.github.com/search/repositories`
- Authentication: Bearer Auth
- Headers:
  - User-Agent
  - Accept: application/vnd.github+json

Code node — Map

Maps GitHub response to minimal JSON (repos list).

Activate the workflow.

Production test

```
curl -X POST https://n8n.your-domain.com/webhook/kuz/github-search   -H 'Content-Type:
application/json'   -d '{"q":"topic:llm
language:python","sort":"stars","per_page":5,"key":"SHARED_SECRET"}'
```

STEP 5 — WordPress Plugin

Plugin structure

```
wp-content/plugins/kuzai-github-search/
├── kuzai-github-search.php
├── kuz-github.js
└── kuz-github.css
```

Plugin activation

- Activate plugin in WordPress admin
- Use shortcode:

```
[kuz_github_search]
```

How It Works (Flow)

1. WordPress form posts to `admin-ajax.php`
2. WordPress calls n8n webhook with shared secret
3. n8n queries GitHub API securely
4. n8n returns sanitized JSON
5. WordPress displays results in a modal

Why This Architecture

- **No secrets in the browser**
- **Secure and compliant GitHub API usage**
- **Scalable and extensible**
- **Excellent UX (modal, no reload)**

Typical Use Cases

- Tech scouting and curation

- AI / Open Source portals
- Documentation or community hubs

Next Improvements

- Add caching (5–10 min) to respect GitHub rate limits
- Add pagination and topic badges
- IP allowlisting on n8n webhooks

Final Note

> Participate in the great project of the universe.
> **One to rule them all.**

#TheKuzNetwork #KUZAI #OpenSource #Automation #n8n

KUZAI.ORG - LIVE AI NEWS FEED - MODAL WINDOW

Open?Source Live AI News Feed

Project: **THE KUZ NETWORK**
Tags: `#AI` `#LocalLLM` `#OpenSource` `#n8n` `#TheKuzNetwork`

Goal

Create a **self-updating AI News feed** on **kuzai.org** using **only open-source tools**.

The system displays the latest Artificial Intelligence headlines from public RSS sources in a **sleek neon widget**, continuously refreshed and SEO-friendly.

Architecture Overview

```
n8n (Scheduler)
   ↓
Python RSS Aggregator
   ↓
Static JSON (ai-news.json)
   ↓
SCP / SSH transfer
   ↓
WordPress Plugin (Frontend Widget)
```

Infrastructure

- **VPS A**: n8n + Python RSS aggregator
- **VPS B**: WordPress (kuzai.org)
- **Apache** serves a static JSON file

- **WordPress plugin** renders the feed

Key Benefits

- 100% open source (Python + n8n + WordPress)
- No paid APIs, no proprietary platforms
- Lightweight: one static JSON file
- Automatic refresh → SEO + user retention
- Easily extendable (filters, tagging, multilingual, OSS focus)

STEP 1 — Debian A: n8n Server & Python Environment

1.1 Create environment and install dependencies

```
mkdir -p /opt/kuznews
cd /opt/kuznews
apt update && apt install -y python3-venv
```

```
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install feedparser html2text requests langdetect
pip freeze > requirements.txt
```

STEP 2 — RSS Feed Configuration

2.1 Feeds list

File: `/opt/kuznews/feeds.txt`

```
https://news.google.com/rss/search?q="artificial+intelligence"+OR+"AI"&hl=en-US&gl=US&ceid=US:en
https://feeds.arstechnica.com/arstechnica/technology-lab
https://www.theverge.com/rss/index.xml
```

STEP 3 — Python Aggregator Script

File: `/opt/kuznews/ai_news_aggregator.py`

```
#!/usr/bin/env python3
import os, sys, time, json, hashlib, requests, feedparser
from langdetect import detect
from pathlib import Path

BASE = Path("/opt/kuznews")
FEEDS = BASE / "feeds.txt"
DB    = BASE / "ai-news-db.json"
OUT   = BASE / "ai-news.json"

def norm(s): return (s or "").strip()
def hash_id(s): return hashlib.sha256(s.encode("utf-8", "ignore")).hexdigest()[:16]

def read_db():
    if DB.exists():
        try: return json.loads(DB.read_text(encoding="utf-8"))
        except: pass
    return {}

def write_db(d):
    DB.write_text(json.dumps(d, ensure_ascii=False, indent=2), encoding="utf-8")

def add_item(db, title, url, published, source):
    if not title or not url: return False
    try:
        if detect(title) != "en": return False
    except: pass
    _id = hash_id(url)
    db[_id] = {
```

```
        "id": _id,
        "title": norm(title),
        "url": url,
        "source": source,
        "published_at": published,
    }
    return True

def fetch_feed(url):
    r = requests.get(url, timeout=15)
    r.raise_for_status()
    return feedparser.parse(r.text)

def main():
    feeds = [l.strip() for l in FEEDS.read_text().splitlines() if l.strip() and not
l.startswith("#")]
    db = read_db()
    for f in feeds:
        try:
            p = fetch_feed(f)
            for e in p.entries[:200]:
                add_item(db, e.get("title"), e.get("link"),
                         e.get("published") or e.get("updated") or "", f)
        except Exception as ex:
            print(f"[WARN] {f}: {ex}", file=sys.stderr)

    items = list(db.values())
    items.sort(key=lambda x: x.get("published_at") or "", reverse=True)

    OUT.write_text(json.dumps({
        "generated_at": time.strftime("%Y-%m-%dT%H:%M:%S%z"),
        "count": len(items),
        "items": items
    }, ensure_ascii=False, indent=2))
    write_db(db)
    print(json.dumps({"total": len(items)}))

if __name__ == "__main__":
    main()
```

STEP 4 — n8n Integration

4.1 Docker Compose update (volume & SSH)

n8n container mounts:

- `/opt/kuznews`
- SSH keys for SCP

4.2 n8n Workflow

Nodes:

1. **Schedule Trigger** — every 15 minutes
2. **Execute Command** — run Python script
3. **Execute Command** — SCP upload to WordPress host

Upload command:

```
scp -i /home/node/.ssh/id_ed25519 /opt/kuznews/ai-news.json
root@91.234.194.49:/var/www/html/kuzai.org/news/ai-news.json
```

STEP 5 — Debian B: WordPress Host

```
mkdir -p /var/www/html/kuzai.org/news
chown -R www-data:www-data /var/www/html/kuzai.org/news
chmod 755 /var/www/html/kuzai.org/news
```

Verification:

```
curl -I https://kuzai.org/news/ai-news.json
```

STEP 6 — WordPress Plugin: “KUZAI AI News”

6.1 Plugin Structure

```
wp-content/plugins/kuzai-ai-news/
├── kuzai-ai-news.php
└── assets/
    ├── js/kuzai-ai-news.js
    └── css/kuzai-ai-news.css
```

6.2 Shortcode

```
[kuz_ai_news limit="9999"]
```

Result

KUZAI.ORG — The Living Pulse of the Open AI Web

- Fully automated
- Continuously updated
- No closed ecosystem
- Transparent, human-scale infrastructure

This project demonstrates that an **independent laboratory** can collect, automate, and publish AI knowledge using **only open technologies**.

© THE KUZ NETWORK

KUZAI.ORG - THE KUZNETWORK KNOWLEDGE SEARCH ENGINE

Detailed Installation Guide (8 Steps) **Includes full code: BookStack indexer, GitHub indexer, PHP proxy, and search UI**

This document describes a complete, reusable installation of a **unified search engine** for a public portal (example: **kuzai.org**) that indexes:

- **BookStack** (wiki) via API + Markdown export
- **GitHub** (repositories + README) via GitHub API and exposes search via:
- **Meilisearch** (backend)
- **PHP proxy** (public gateway without exposing keys)
- **Standalone** `search.html` (UI)

The process is structured in **8 steps** and is designed for third-party reuse.

Reference topology (adapt as needed)

- **VPS A (Docker + indexing)**: `83.228.192.116`
  - Meilisearch listens on `:7700` (firewall-restricted)
  - Python indexers run on the host
- **VPS B (Public Web)**: `91.234.194.49`
  - `kuzai.org` (WordPress + static files)
  - `kuz-search-api.php` proxy endpoint
- **BookStack**: `https://wiki.kuzlab.org`
- **GitHub**: `https://github.com/kusanagi8200`

Security model (recommended)

- Port `7700` is **not public**.
- Firewall rule: allow `TCP/7700` **from** VPS B only (public web server IP) to VPS A.
- The **Meilisearch master key stays on VPS A**.
- VPS B stores only a **search-only Meilisearch key**.

STEP 1 — Deploy Meilisearch (VPS A)

1.1 Directory

```
mkdir -p /opt/kuz-search/meilisearch
cd /opt/kuz-search/meilisearch
```

1.2 `docker-compose.yml`

Create `/opt/kuz-search/meilisearch/docker-compose.yml`:

```
services:
  meilisearch:
    image: getmeili/meilisearch:latest
    container_name: kuz-meilisearch
    environment:
      MEILI_MASTER_KEY: "CHANGE_ME_LONG_RANDOM_KEY"
      MEILI_ENV: "production"
    volumes:
      - meili_data:/meili_data
    ports:
      - "0.0.0.0:7700:7700"
    restart: unless-stopped

volumes:
  meili_data:
```

1.3 Start

```
docker compose up -d
docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}'
```

1.4 Verify

```
curl -sS http://127.0.0.1:7700/health
```

Expected:

```
{"status":"available"}
```

STEP 2 — Create the index and core settings (VPS A)

2.1 Export master key for current session

```
export MEILI_KEY='CHANGE_ME_LONG_RANDOM_KEY'
```

2.2 Create index

```
curl -sS -X POST 'http://127.0.0.1:7700/indexes' \
  -H "Authorization: Bearer ${MEILI_KEY}" \
  -H 'Content-Type: application/json' \
  --data '{"uid":"kuz_universal","primaryKey":"id"}'
```

2.3 Apply settings (includes typoTolerance object)

```
curl -sS -X PATCH 'http://127.0.0.1:7700/indexes/kuz_universal/settings' \
  -H "Authorization: Bearer ${MEILI_KEY}" \
  -H 'Content-Type: application/json' \
  --data '{
    "searchableAttributes": ["title","content","repo","path","tags"],
    "filterableAttributes":
["source","type","space","repo","book_id","book_slug","chapter_id"],
    "sortableAttributes": ["updated_at"],
    "displayedAttributes":
["id","source","type","space","repo","book_id","book_slug","chapter_id","title","url","path","
tags","updated_at","content"],
    "typoTolerance": {
      "enabled": true,
      "minWordSizeForTypos": { "oneTypo": 4, "twoTypos": 7 }
    }
  }'
```

2.4 Verify settings

```
curl -sS 'http://127.0.0.1:7700/indexes/kuz_universal/settings' \
  -H "Authorization: Bearer ${MEILI_KEY}" | head -n 80
```

STEP 3 — Validate BookStack API + exports (VPS A)

3.1 Set BookStack variables (session)

```
export BOOKSTACK_URL='https://wiki.kuzlab.org'
export BOOKSTACK_TOKEN_ID='XXXX'
export BOOKSTACK_TOKEN_SECRET='YYYY'
```

3.2 Test page list

```
curl -sS "${BOOKSTACK_URL}/api/pages?count=1" \
  -H "Authorization: Token ${BOOKSTACK_TOKEN_ID}:${BOOKSTACK_TOKEN_SECRET}" \
  -H "Accept: application/json" | head -n 120
```

3.3 Test Markdown export (example page id)

```
export PAGE_ID=1
curl -sS -i "${BOOKSTACK_URL}/api/pages/${PAGE_ID}/export/markdown" \
  -H "Authorization: Token ${BOOKSTACK_TOKEN_ID}:${BOOKSTACK_TOKEN_SECRET}" \
  -H "Accept: text/plain" | head -n 40
```

Expected:

- `HTTP/1.1 200 OK`
- Markdown content in body

STEP 4 — BookStack ? Meilisearch indexer (full code) (VPS A)

4.1 Python environment

```
apt-get update
apt-get install -y python3 python3-venv python3-pip
mkdir -p /opt/kuz-search/indexers
python3 -m venv /opt/kuz-search/venv
/opt/kuz-search/venv/bin/pip install --upgrade pip
/opt/kuz-search/venv/bin/pip install requests
```

4.2 Create `/opt/kuz-search/indexers/bookstack_to_meili.py`

```
###!/usr/bin/env python3
import os
import time
import hashlib
import requests

BOOKSTACK_URL = os.environ["BOOKSTACK_URL"].rstrip("/")
TOKEN_ID = os.environ["BOOKSTACK_TOKEN_ID"]
TOKEN_SECRET = os.environ["BOOKSTACK_TOKEN_SECRET"]

MEILI_URL = os.environ.get("MEILI_URL", "http://127.0.0.1:7700").rstrip("/")
MEILI_KEY = os.environ["MEILI_KEY"]
INDEX = os.environ.get("MEILI_INDEX", "kuz_universal")

### Test mode: index only the first page if LIMIT_ONE=1
LIMIT_ONE = os.environ.get("LIMIT_ONE", "0") == "1"



def bs_headers_json():
    return {
        "Authorization": f"Token {TOKEN_ID}:{TOKEN_SECRET}",
        "Accept": "application/json",
    }



def bs_headers_download():
    return {
        "Authorization": f"Token {TOKEN_ID}:{TOKEN_SECRET}",
        "Accept": "application/octet-stream",
    }



def meili_headers():
    return {
        "Authorization": f"Bearer {MEILI_KEY}",
        "Content-Type": "application/json",
    }
```

```
def sid(value: str) -> str:
    return hashlib.sha1(value.encode("utf-8")).hexdigest()


def list_pages():
    pages = []
    offset = 0
    count = 100

    while True:
        r = requests.get(
            f"{BOOKSTACK_URL}/api/pages",
            headers=bs_headers_json(),
            params={"count": count, "offset": offset},
            timeout=30,
        )
        r.raise_for_status()
        data = r.json()
        chunk = data.get("data", [])
        pages.extend(chunk)

        if len(chunk) < count:
            break

        offset += count
        time.sleep(0.2)

    return pages


def export_markdown(page_id: int) -> str:
    r = requests.get(
        f"{BOOKSTACK_URL}/api/pages/{page_id}/export/markdown",
        headers=bs_headers_download(),
        timeout=60,
    )
    r.raise_for_status()
    return r.text
```

```
def push_docs(docs):
    r = requests.post(
        f"{MEILI_URL}/indexes/{INDEX}/documents",
        headers=meili_headers(),
        json=docs,
        timeout=60,
    )
    r.raise_for_status()


def main():
    pages = list_pages()
    if not pages:
        print("No pages found.")
        return

    batch = []
    total = len(pages)

    for i, p in enumerate(pages, start=1):
        page_id = p["id"]
        title = p.get("name", f"Page {page_id}")
        slug = p.get("slug", "")
        updated = p.get("updated_at", "")

        # BookStack canonical URL pattern
        url = (
            f"{BOOKSTACK_URL}/books/{p.get('book_slug','')}/page/{slug}"
            if slug
            else f"{BOOKSTACK_URL}/"
        )

        md = export_markdown(page_id)

        doc = {
            "id": sid(f"bookstack:{page_id}:{url}"),
            "source": "bookstack",
            "type": "page",
            "space": "wiki.kuzlab.org",
```

```
            "repo": "",
            "book_id": p.get("book_id", 0),
            "book_slug": p.get("book_slug", ""),
            "chapter_id": p.get("chapter_id", 0),
            "title": title,
            "url": url,
            "path": slug,
            "tags": [],
            "updated_at": updated,
            "content": md,
        }

        batch.append(doc)

        # Send in batches
        if len(batch) >= 20:
            push_docs(batch)
            batch = []

        print(f"[{i}/{total}] queued page_id={page_id} title={title}")
        time.sleep(0.1)

        if LIMIT_ONE:
            break

    if batch:
        push_docs(batch)

    print("DONE")


if __name__ == "__main__":
    main()
```

Permissions:

```
chmod +x /opt/kuz-search/indexers/bookstack_to_meili.py
```

4.3 Run (test mode, 1 page)

```
export LIMIT_ONE=1
/opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/bookstack_to_meili.py
```

4.4 Verify in Meilisearch

```
curl -sS 'http://127.0.0.1:7700/indexes/kuz_universal/search' \
  -H "Authorization: Bearer ${MEILI_KEY}" \
  -H 'Content-Type: application/json' \
  --data '{"q":"WHITE BOOK","limit":5,"filter":"source = bookstack"}' | head -n 160
```

4.5 Run full import

```
unset LIMIT_ONE
/opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/bookstack_to_meili.py
```

Stats:

```
curl -sS 'http://127.0.0.1:7700/indexes/kuz_universal/stats' \
  -H "Authorization: Bearer ${MEILI_KEY}" | head -n 120
```

STEP 5 — GitHub ? Meilisearch indexer (full code) (VPS A)

5.1 Export GitHub variables

```
export GITHUB_USER='kusanagi8200'
export GITHUB_TOKEN='ZZZZ'
```

5.2 Create `/opt/kuz-search/indexers/github_to_meili.py`

```
###!/usr/bin/env python3
import os
import time
import hashlib
import base64
import requests

GITHUB_USER = os.environ.get("GITHUB_USER", "kusanagi8200")
GITHUB_TOKEN = os.environ.get("GITHUB_TOKEN", "")
```

```
MEILI_URL = os.environ.get("MEILI_URL", "http://127.0.0.1:7700").rstrip("/")
MEILI_KEY = os.environ["MEILI_KEY"]
INDEX = os.environ.get("MEILI_INDEX", "kuz_universal")

### Test mode: index only the first repo if LIMIT_ONE=1
LIMIT_ONE = os.environ.get("LIMIT_ONE", "0") == "1"



def sid(value: str) -> str:
    return hashlib.sha1(value.encode("utf-8")).hexdigest()



def gh_headers():
    h = {"Accept": "application/vnd.github+json"}
    if GITHUB_TOKEN:
        h["Authorization"] = f"Bearer {GITHUB_TOKEN}"
    return h



def meili_headers():
    return {"Authorization": f"Bearer {MEILI_KEY}", "Content-Type": "application/json"}



def list_repos():
    repos = []
    page = 1

    while True:
        r = requests.get(
            f"https://api.github.com/users/{GITHUB_USER}/repos",
            headers=gh_headers(),
            params={"per_page": 100, "page": page, "sort": "updated"},
            timeout=30,
        )
        r.raise_for_status()
        chunk = r.json()
        if not chunk:
            break
```

```
        repos.extend(chunk)
        page += 1
        time.sleep(0.2)

    return repos


def get_readme(owner, repo):
    r = requests.get(
        f"https://api.github.com/repos/{owner}/{repo}/readme",
        headers=gh_headers(),
        timeout=30,
    )
    if r.status_code == 404:
        return None
    r.raise_for_status()
    data = r.json()
    content_b64 = data.get("content", "")
    if not content_b64:
        return None

    raw = base64.b64decode(content_b64).decode("utf-8", errors="replace")
    return raw, data.get("html_url")


def push_docs(docs):
    if not docs:
        return
    r = requests.post(
        f"{MEILI_URL}/indexes/{INDEX}/documents",
        headers=meili_headers(),
        json=docs,
        timeout=60,
    )
    r.raise_for_status()


def main():
    repos = list_repos()
    if not repos:
```

```
        print("No repos found.")
        return

    out = []
    total = len(repos)

    for i, repo in enumerate(repos, start=1):
        name = repo["name"]
        html_url = repo["html_url"]
        updated = repo.get("updated_at", "")
        owner = repo["owner"]["login"]
        desc = repo.get("description") or ""

        # Repo document
        out.append(
            {
                "id": sid(f"github:repo:{html_url}"),
                "source": "github",
                "type": "repo",
                "space": "github.com",
                "repo": name,
                "title": name,
                "url": html_url,
                "path": "",
                "tags": [],
                "updated_at": updated,
                "content": desc,
            }
        )

        # README document
        readme = get_readme(owner, name)
        if readme:
            readme_text, readme_url = readme
            url = readme_url or html_url
            out.append(
                {
                    "id": sid(f"github:readme:{url}"),
                    "source": "github",
                    "type": "readme",
```

```
                    "space": "github.com",
                    "repo": name,
                    "title": f"{name} — README",
                    "url": url,
                    "path": "README",
                    "tags": [],
                    "updated_at": updated,
                    "content": readme_text,
                }
            )

        if len(out) >= 200:
            push_docs(out)
            out = []

        print(f"[{i}/{total}] queued repo={name}")
        time.sleep(0.1)

        if LIMIT_ONE:
            break

    if out:
        push_docs(out)

    print("DONE")


if __name__ == "__main__":
    main()
```

Permissions:

```
chmod +x /opt/kuz-search/indexers/github_to_meili.py
```

5.3 Run test mode (1 repo)

```
export LIMIT_ONE=1
/opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/github_to_meili.py
```

5.4 Run full import

```
unset LIMIT_ONE
/opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/github_to_meili.py
```

Verify:

```
curl -sS 'http://127.0.0.1:7700/indexes/kuz_universal/stats' \
  -H "Authorization: Bearer ${MEILI_KEY}" | head -n 120
```

STEP 6 — Create a Meilisearch search-only key (VPS A)

The public web server must never use the master key.

```
export MEILI_MASTER_KEY='CHANGE_ME_LONG_RANDOM_KEY'


curl -sS -X POST 'http://127.0.0.1:7700/keys' \
  -H "Authorization: Bearer ${MEILI_MASTER_KEY}" \
  -H 'Content-Type: application/json' \
  --data '{
    "description": "kuzai.org (web) search-only key",
    "actions": ["search"],
    "indexes": ["kuz_universal"],
    "expiresAt": null
  }'
```

Store the returned `key`.

Verify keys (optional):

```
curl -sS 'http://127.0.0.1:7700/keys' \
  -H "Authorization: Bearer ${MEILI_MASTER_KEY}" | head -n 200
```

STEP 7 — Deploy the PHP proxy on the public web server (VPS B)

7.1 Create `/etc/kuz-search/meili.env` (VPS B)

```
mkdir -p /etc/kuz-search
chmod 750 /etc/kuz-search
chown root:www-data /etc/kuz-search

cat > /etc/kuz-search/meili.env <<'ENV'
MEILI_URL=http://83.228.192.116:7700
MEILI_INDEX=kuz_universal
MEILI_SEARCH_KEY=PASTE_SEARCH_KEY_HERE
ENV

chown root:www-data /etc/kuz-search/meili.env
chmod 640 /etc/kuz-search/meili.env
```

7.2 Create `/var/www/html/kuzai.org/kuz-search-api.php` (VPS B)

```
<?php
declare(strict_types=1);

header('Content-Type: application/json; charset=utf-8');

$q = trim((string)($_GET['q'] ?? ''));
$source = trim((string)($_GET['source'] ?? ''));     // bookstack|github
$type   = trim((string)($_GET['type'] ?? ''));       // page|repo|readme
$repo   = trim((string)($_GET['repo'] ?? ''));       // repo name
$book   = trim((string)($_GET['book_slug'] ?? ''));  // book slug
$limit  = (int)($_GET['limit'] ?? 10);
$limit  = max(1, min(20, $limit));

if (strlen($q) > 200) {
  http_response_code(400);
  echo json_encode(['error' => 'query_too_long']);
  exit;
}

$envFile = '/etc/kuz-search/meili.env';
if (!is_readable($envFile)) {
  http_response_code(500);
  echo json_encode(['error' => 'config_missing']);
  exit;
```

```
}

$env = [];
foreach (file($envFile, FILE_IGNORE_NEW_LINES | FILE_SKIP_EMPTY_LINES) as $line) {
  if ($line === '' || $line[0] === '#') continue;
  $parts = explode('=', $line, 2);
  if (count($parts) !== 2) continue;
  $k = trim($parts[0]);
  $v = trim($parts[1]);
  $env[$k] = $v;
}

$meiliUrl = rtrim($env['MEILI_URL'] ?? '', '/');
$index    = $env['MEILI_INDEX'] ?? 'kuz_universal';
$apiKey   = $env['MEILI_SEARCH_KEY'] ?? '';

if ($meiliUrl === '' || $apiKey === '') {
  http_response_code(500);
  echo json_encode(['error' => 'config_invalid']);
  exit;
}

$filters = [];
$allowSource = ['bookstack', 'github'];
$allowType   = ['page', 'repo', 'readme'];

if ($source !== '' && in_array($source, $allowSource, true)) {
  $filters[] = 'source = "'.addslashes($source).'"';
}
if ($type !== '' && in_array($type, $allowType, true)) {
  $filters[] = 'type = "'.addslashes($type).'"';
}
if ($repo !== '') {
  if (!preg_match('/^[A-Za-z0-9._-]{1,100}$/', $repo)) {
    http_response_code(400);
    echo json_encode(['error' => 'invalid_repo']);
    exit;
  }
  $filters[] = 'repo = "'.addslashes($repo).'"';
}
```

```
if ($book !== '') {
  if (!preg_match('/^[A-Za-z0-9._-]{1,120}$/', $book)) {
    http_response_code(400);
    echo json_encode(['error' => 'invalid_book_slug']);
    exit;
  }
  $filters[] = 'book_slug = "'.addslashes($book).'"';
}

$payload = [
  'q' => $q,
  'limit' => $limit,
  'attributesToRetrieve' => [
    'id','source','type','space','repo','book_slug','book_id','chapter_id',
    'title','url','path','tags','updated_at'
  ],
  'attributesToHighlight' => ['title','content'],
  'highlightPreTag' => '<mark>',
  'highlightPostTag' => '</mark>',
];

if ($filters) $payload['filter'] = implode(' AND ', $filters);

$ch = curl_init();
curl_setopt_array($ch, [
  CURLOPT_URL => $meiliUrl . '/indexes/' . rawurlencode($index) . '/search',
  CURLOPT_RETURNTRANSFER => true,
  CURLOPT_POST => true,
  CURLOPT_HTTPHEADER => [
    'Authorization: Bearer ' . $apiKey,
    'Content-Type: application/json',
  ],
  CURLOPT_POSTFIELDS => json_encode($payload),
  CURLOPT_TIMEOUT => 10,
]);

$res = curl_exec($ch);
$err = curl_error($ch);
$code = (int)curl_getinfo($ch, CURLINFO_HTTP_CODE);
curl_close($ch);
```

```
if ($res === false) {
  http_response_code(502);
  echo json_encode(['error' => 'meili_unreachable', 'detail' => $err]);
  exit;
}

http_response_code($code ?: 200);
echo $res;
```

Permissions:

```
chmod 644 /var/www/html/kuzai.org/kuz-search-api.php
```

7.3 Verify from VPS B

```
curl -sS http://83.228.192.116:7700/health
curl -sS 'https://kuzai.org/kuz-search-api.php?q=INFRASTRUCTURE&source=bookstack&limit=5' |
head -n 160
```

STEP 8 — Deploy `search.html` UI + automate reindexing

8.1 Deploy `/var/www/html/kuzai.org/search.html` (VPS B)

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>KUZAI — Universal Search</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
:root{
  --bg:#050b14; --fg:#e6f1ff; --accent:#33ccff; --muted:#8aa0b6;
}
body{
  margin:0; font-family:system-ui,-apple-system,Segoe UI,Roboto,sans-serif;
  background:radial-gradient(1200px 600px at 50% -20%, #0a1b33, var(--bg));
  color:var(--fg);
```

```
}
.container{ max-width:980px; margin:80px auto; padding:0 16px; }
h1{ font-size:28px; letter-spacing:.4px; margin:0 0 12px; }
.searchbox{ display:flex; gap:8px; margin:16px 0 12px; }
.searchbox input{
  flex:1; padding:12px 14px; border-radius:10px; border:1px solid #1b3557;
  background:#071427; color:var(--fg);
}
.searchbox button{
  padding:12px 18px; border-radius:10px; border:1px solid #1b3557;
  background:linear-gradient(180deg, #0f6fb2, #0a4e86); color:#fff; cursor:pointer;
}
.scopes{ display:flex; gap:8px; margin-bottom:18px; }
.scope{
  padding:6px 10px; border-radius:999px; border:1px solid #1b3557;
  color:var(--muted); cursor:pointer; user-select:none;
}
.scope.active{
  color:#fff; border-color:var(--accent);
  box-shadow:0 0 0 1px rgba(51,204,255,.3) inset;
}
.results{ margin-top:18px; }
.card{
  padding:14px 16px; border-radius:14px; border:1px solid #1b3557;
  background:#071427; margin-bottom:10px;
}
.card .meta{ font-size:12px; color:var(--muted); margin-bottom:4px; }
.card a{ color:var(--accent); text-decoration:none; }
mark{ background:rgba(51,204,255,.25); color:#fff; padding:0 .15em; border-radius:3px; }
</style>
</head>
<body>
<div class="container">
  <h1>KUZAI — Universal Search</h1>

  <div class="searchbox">
    <input id="q" placeholder="Search documentation, code, projects…" autofocus>
    <button id="btn">Search</button>
  </div>
```

```
  <div class="scopes">
    <div class="scope active" data-scope="">ALL</div>
    <div class="scope" data-scope="bookstack">WIKI</div>
    <div class="scope" data-scope="github">GITHUB</div>
  </div>

  <div id="results" class="results"></div>
</div>

<script>
let scope = "";

function setActive(el){
  document.querySelectorAll('.scope').forEach(s => s.classList.remove('active'));
  el.classList.add('active');
  scope = el.dataset.scope || "";
  runSearch();
}

document.querySelectorAll('.scope').forEach(el => {
  el.addEventListener('click', () => setActive(el));
});

async function runSearch(){
  const q = document.getElementById('q').value.trim();
  const out = document.getElementById('results');
  if(!q){ out.innerHTML = ""; return; }

  const params = new URLSearchParams({ q, limit: "10" });
  if(scope) params.set('source', scope);

  const res = await fetch('/kuz-search-api.php?' + params.toString());
  const data = await res.json();

  out.innerHTML = "";
  (data.hits || []).forEach(h => {
    const card = document.createElement('div');
    card.className = 'card';
    card.innerHTML = `
      <div class="meta">${String(h.source||'').toUpperCase()} • ${h.type||''}</div>
```

```
      <a href="${h.url}" target="_blank" rel="noopener"><strong>${h.title}</strong></a>
    `;
    out.appendChild(card);
  });
}

document.getElementById('btn').addEventListener('click', runSearch);
document.getElementById('q').addEventListener('keydown', (e) => { if(e.key === 'Enter')
runSearch(); });
</script>
</body>
</html>
```

Permissions:

```
chmod 644 /var/www/html/kuzai.org/search.html
```

Verify:

```
curl -sS 'https://kuzai.org/kuz-search-api.php?q=KUZ&limit=5' | head -n 200
```

8.2 Automation (VPS A): persistent env + cron reindexing

Create `/etc/kuz-search/indexers.env` (VPS A)

```
mkdir -p /etc/kuz-search
chmod 700 /etc/kuz-search

cat > /etc/kuz-search/indexers.env <<'ENV'
### BookStack
BOOKSTACK_URL=https://wiki.kuzlab.org
BOOKSTACK_TOKEN_ID=XXXX
BOOKSTACK_TOKEN_SECRET=YYYY

### GitHub
GITHUB_USER=kusanagi8200
GITHUB_TOKEN=ZZZZ

### Meilisearch local (MASTER key)
MEILI_URL=http://127.0.0.1:7700
```

```
MEILI_INDEX=kuz_universal
MEILI_KEY=CHANGE_ME_LONG_RANDOM_KEY
ENV

chmod 600 /etc/kuz-search/indexers.env
```

Install and enable cron (if needed)

```
apt-get update
apt-get install -y cron
systemctl enable cron
systemctl start cron
systemctl status cron --no-pager
```

Create `/etc/cron.d/kuz-search` (example: both every 6 hours)

```
mkdir -p /var/log/kuz-search
chmod 755 /var/log/kuz-search

cat > /etc/cron.d/kuz-search <<'CRON'
SHELL=/bin/bash
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

### BookStack → Meilisearch (every 6 hours)
0 */6 * * * root set -a; source /etc/kuz-search/indexers.env; set +a; /opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/bookstack_to_meili.py >> /var/log/kuz-search/bookstack.log 2>&1

### GitHub → Meilisearch (every 6 hours)
15 */6 * * * root set -a; source /etc/kuz-search/indexers.env; set +a; /opt/kuz-search/venv/bin/python /opt/kuz-search/indexers/github_to_meili.py >> /var/log/kuz-search/github.log 2>&1
CRON

chmod 644 /etc/cron.d/kuz-search
```

Verify:

```
journalctl -u cron --no-pager | tail -n 60
tail -n 60 /var/log/kuz-search/bookstack.log
tail -n 60 /var/log/kuz-search/github.log
```

Final verification checklist

VPS A

```
curl -sS http://127.0.0.1:7700/health
curl -sS 'http://127.0.0.1:7700/indexes/kuz_universal/stats' -H "Authorization: Bearer
${MEILI_KEY}"
```

VPS B

```
curl -sS http://83.228.192.116:7700/health
curl -sS 'https://kuzai.org/kuz-search-api.php?q=INFRASTRUCTURE&source=bookstack&limit=5' |
head -n 200
```

Browser:

- `https://kuzai.org/search.html`
- Search terms present in BookStack and GitHub README

Appendix — Generate a strong master key

```
python3 - <<'PY'
import secrets
print(secrets.token_urlsafe(48))
PY
```

KUZAI.ORG - KUZ-METRIKS

Overview

This document describes the full installation and integration process for **Metriks** (tracking endpoint + MySQL storage + web dashboard) as deployed on:

- `metriks.kuzlab.org` (collector + dashboard)
- `kuzai.org` (SID=1)
- `mob.kuzai.org` (SID=2)
- `radio.kuzai.org` (SID=3)

Referrer tracking has been **removed** (no referrer collection/display).

Installation action list

- Provision MySQL database `metrics` and a dedicated user.
- Create core tables: `sites`, `daily_site`, `daily_page`, `unique_seen`.
- Register each tracked site (SID/host/label) with a 32-byte secret.
- Deploy collector webroot to `/var/www/metriks`.
- Configure Apache vhosts for `metriks.kuzlab.org` (HTTP→HTTPS, PHP enabled).
- Route `/i.gif` to `/i.php` (PHP collector).
- Deploy `dashboard.php` and `index.php` redirect.
- Deploy KUZAI theme CSS and background image assets.
- Deploy `beacon.php` on each tracked site (kuzai/mob/radio) with correct constants.
- Inject beacon image/JS in WordPress and AzuraCast public pages.
- Verify collector headers with `curl -I` (`X-Metriks-Handler: php`).
- Verify end-to-end metrics with a signed test request and DB queries.

0) Files and directories

Project layout (collector host):

```
/var/www/metriks
├── config
│   └── config.php
├── public
│   ├── assets
│   │   ├── css
│   │   │   └── kuzai-theme.css
│   │   └── img
│   │       └── FOND-GITHUB-SEARCH2.png
```

```
│   ├── dashboard.php
│   ├── i.gif
│   ├── i.php
│   └── index.php
└── src
    ├── Auth.php
    ├── Db.php
    └── Metrics.php
```

---

1) MySQL database and user

Create the DB and a dedicated user.

```
mysql -e "CREATE DATABASE IF NOT EXISTS metrics CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;"

mysql -e "CREATE USER IF NOT EXISTS 'metrics_user'@'localhost' IDENTIFIED BY '$PW#';"
mysql -e "GRANT ALL PRIVILEGES ON metrics.* TO 'metrics_user'@'localhost';"
mysql -e "FLUSH PRIVILEGES;"
```

Verification:

```
mysql -e "SHOW DATABASES LIKE 'metrics';"
mysql -e "SHOW GRANTS FOR 'metrics_user'@'localhost';"
```

---

2) Tables

Core tables used by the collector + dashboard.

```
mysql metrics <<'SQL'
CREATE TABLE IF NOT EXISTS sites (
  id INT NOT NULL AUTO_INCREMENT,
  host VARCHAR(255) NOT NULL,
  label VARCHAR(255) NULL,
  secret BINARY(32) NOT NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uniq_host (host)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE IF NOT EXISTS daily_site (
```

```
  site_id INT NOT NULL,
  day DATE NOT NULL,
  hits INT NOT NULL DEFAULT 0,
  uniques INT NOT NULL DEFAULT 0,
  PRIMARY KEY (site_id, day),
  CONSTRAINT fk_daily_site_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE IF NOT EXISTS daily_page (
  site_id INT NOT NULL,
  day DATE NOT NULL,
  path VARCHAR(1024) NOT NULL,
  hits INT NOT NULL DEFAULT 0,
  uniques INT NOT NULL DEFAULT 0,
  PRIMARY KEY (site_id, day, path(191)),
  CONSTRAINT fk_daily_page_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- unique_seen stores one row per visitor+page per day.
-- Columns shown here match the current collector/dashboard usage.
CREATE TABLE IF NOT EXISTS unique_seen (
  site_id INT NOT NULL,
  day DATE NOT NULL,
  scope ENUM('site','page') NOT NULL,
  path VARCHAR(1024) NOT NULL,
  visitor_hash BINARY(32) NOT NULL,
  last_seen_at DATETIME NOT NULL,
  ip VARBINARY(16) NULL,
  ua VARCHAR(512) NOT NULL,
  hits INT NOT NULL DEFAULT 1,
  PRIMARY KEY (site_id, day, scope, visitor_hash, path(191)),
  CONSTRAINT fk_unique_seen_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
SQL
```

Verification:

```
mysql -N metrics -e "SHOW TABLES;"
mysql -N metrics -e "SHOW CREATE TABLE unique_seen\G" | sed -n '1,160p'
```

3) Sites registration (SIDs + secrets)

Insert the sites and generate a 32-byte secret per site.

Example (generate secrets):

```
# 32 random bytes (hex)
openssl rand -hex 32
```

Insert (example values):

```
mysql metrics <<'SQL'
INSERT INTO sites (id, host, label, secret) VALUES
  (1, 'kuzai.org',       'KUZAI',       UNHEX('<HEX_64_CHARS_FOR_SID1>')),
  (2, 'mob.kuzai.org',   'KUZAI-MOB',   UNHEX('$SECRET')),
  (3, 'radio.kuzai.org', 'KUZAI-RADIO', UNHEX('$SECRET<REST_OF_HEX>'))
ON DUPLICATE KEY UPDATE host=VALUES(host), label=VALUES(label), secret=VALUES(secret);
SQL
```

Verification:

```
mysql -N metrics -e "SELECT id, host, label, LENGTH(secret) AS secret_len FROM sites ORDER BY id;"
```

---

4) Apache vhost for metriks.kuzlab.org

Collector and dashboard are served from `/var/www/metriks/public`.

Key requirements:

- HTTPS enabled (Let’s Encrypt)
- `/i.gif` must be handled by PHP (`/i.php`) so that it increments DB counters

Example check (vhosts + modules):

```
apache2ctl -S | grep -E "metriks\.kuzlab\.org|:80|:443" -n || true
apache2ctl -M | grep -E 'rewrite|headers|php|proxy_fcgi' || true
```

Verification (endpoint must show `X-Metriks-Handler: php`):

```
curl -I "https://metriks.kuzlab.org/i.php?x=$(date +%s)" | head -n 40
curl -I "https://metriks.kuzlab.org/i.gif?x=$(date +%s)" | head -n 40
```

---

5) Beacon integration (kuzai.org / mob.kuzai.org / radio.kuzai.org)

Each site exposes a local `beacon.php` that:

- computes a signed token
- redirects to `https://metriks.kuzlab.org/i.gif?...`

Important: referrer is not used. The path is passed explicitly using `?p=<path>`.

5.1) beacon.php (template)

Deploy this file on each site as `/beacon.php` and set `SID` and `SECRET_HEX`.

```
<?php
declare(strict_types=1);

// ===== METRIKS BEACON (NO REFERRER) =====
// Per-site constants
const SID = 0; // TODO: set 1/2/3
const SECRET_HEX = ''; // TODO: 64 hex chars
const METRICS_BASE = 'https://metriks.kuzlab.org/i.gif';

function clampPath(string $p): string {
    $p = trim($p);
    if ($p === '') return '/';
    if ($p[0] !== '/') $p = '/' . $p;
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
    if (mb_strlen($p, 'UTF-8') > 1024) $p = mb_substr($p, 0, 1024, 'UTF-8');
    return $p;
}

$path = clampPath((string)($_GET['p'] ?? '/'));
$ts   = (string)time();

$secret = hex2bin(SECRET_HEX);
if ($secret === false) {
    http_response_code(500);
    exit;
}

$msg = SID . "\n" . $path . "\n" . $ts;
$tok = hash_hmac('sha256', $msg, $secret, false);
```

```
$url = METRICS_BASE
    . '?sid=' . rawurlencode((string)SID)
    . '&p='   . rawurlencode($path)
    . '&ts='  . rawurlencode($ts)
    . '&t='   . rawurlencode($tok);

header('Location: ' . $url, true, 302);
exit;
```

5.2) Set constants per site

- `kuzai.org`:
  - `const SID = 1;`
  - `const SECRET_HEX = '<HEX_FOR_SID1>';`
- `mob.kuzai.org`:
  - `const SID = 2;`
  - `const SECRET_HEX = '$SECRET';`
- `radio.kuzai.org`:
  - `const SID = 3;`
  - `const SECRET_HEX = '$SECRET<REST_OF_HEX>';`

5.3) Front-end injection

WordPress (theme): insert a hidden image with the current path.

```
<img src="/beacon.php?p=/" style="display:none" alt="">
<script>
  (function(){
    var i = document.createElement('img');
    i.src = '/beacon.php?p=' + encodeURIComponent(location.pathname) + '&x=' + Date.now();
    i.alt = '';
    i.style.display = 'none';
    document.body.appendChild(i);
  })();
</script>
```

AzuraCast (Public Pages → Custom JS):

```
// === METRIKS BEACON (radio.kuzai.org) ===
(function () {
  if (document.getElementById('metriks-beacon')) return;
```

```
  const img = document.createElement('img');
  img.id = 'metriks-beacon';
  img.alt = '';
  img.width = 1;
  img.height = 1;
  img.style.position = 'absolute';
  img.style.left = '-9999px';
  img.style.width = '1px';
  img.style.height = '1px';
  img.referrerPolicy = 'strict-origin-when-cross-origin';

  const p = encodeURIComponent(location.pathname || '/');
  img.src = 'https://radio.kuzai.org/beacon.php?p=' + p + '&x=' + Date.now();

  const target = document.body || document.head || document.documentElement;
  if (target) target.appendChild(img);
})();
```

Verification (should return a 302 to metriks i.gif):

```
curl -I "https://mob.kuzai.org/beacon.php?p=%2F&x=$(date +%s)" | head -n 25
curl -I "https://radio.kuzai.org/beacon.php?p=%2F&x=$(date +%s)" | head -n 25
```

---

6) Manual collector test (direct hit)

Generate a valid token and call the collector.

```
set -euo pipefail

SID=1
P="/manual-test"
TS="$(date +%s)"
SECRET_HEX="$(mysql -N metrics -e "SELECT HEX(secret) FROM sites WHERE id=${SID};")"

TOKEN="$(SID="$SID" P="$P" TS="$TS" SECRET_HEX="$SECRET_HEX" php -r '
$sid=(int)getenv("SID");
$p=getenv("P");
$ts=getenv("TS");
$secret=hex2bin(getenv("SECRET_HEX"));
echo hash_hmac("sha256", $sid."\n".$p."\n".$ts, $secret, false);
```

```
')"

P_ENC="$(P="$P" php -r 'echo rawurlencode(getenv("P"));')"

curl -s -o /dev/null
"https://metriks.kuzlab.org/i.gif?sid=${SID}&p=${P_ENC}&ts=${TS}&t=${TOKEN}"

mysql -N metrics -e "SELECT site_id, day, hits, uniques FROM daily_site WHERE site_id=${SID}
AND day=CURDATE();"
mysql -N metrics -e "SELECT path, hits, uniques FROM daily_page WHERE site_id=${SID} AND
day=CURDATE() ORDER BY hits DESC LIMIT 10;"
mysql -N metrics -e "SELECT COUNT(*) FROM unique_seen;"
```

---

7) Web app (dashboard)

- `/dashboard.php`: UI (KUZAI theme)
- `/index.php`: redirect to the dashboard

Verification:

```
curl -I https://metriks.kuzlab.org/dashboard.php | head -n 25
curl -I https://metriks.kuzlab.org/ | head -n 20
```

---

8) Deploy/Update application files

Run these commands on the collector host to deploy the current code.

```
set -euo pipefail

# Backup current files
cp -a /var/www/metriks/public/i.php /var/www/metriks/public/i.php.bk.$(date +%s) 2>/dev/null
|| true
cp -a /var/www/metriks/public/dashboard.php /var/www/metriks/public/dashboard.php.bk.$(date
+%s) 2>/dev/null || true
cp -a /var/www/metriks/public/assets/css/kuzai-theme.css
/var/www/metriks/public/assets/css/kuzai-theme.css.bk.$(date +%s) 2>/dev/null || true

# Deploy i.php
cat > /var/www/metriks/public/i.php <<'PHP'
<?php
```

```php
declare(strict_types=1);

/*
  i.php — image endpoint (tracking)
  - Validates HMAC token (sid\npath\nts)
  - Updates daily_site / daily_page / unique_seen
  - Referrer tracking: DISABLED
  - Outputs a 1x1 GIF
*/

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function outGif(): void {
    header('X-Metriks-Handler: php');
    header('Cache-Control: no-store, no-cache, must-revalidate, max-age=0');
    header('Pragma: no-cache');
    header('Content-Type: image/gif');

    // 1x1 transparent gif
    echo base64_decode('R0lGODlhAQABAPAAAAAAAAAAACH5BAEAAAAALAAAAAABAAEAAAICRAEAOw==');
    exit;
}

function getClientIp(): ?string {
    // Prefer first IP from X-Forwarded-For if present (proxy)
    if (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
        $parts = explode(',', (string)$_SERVER['HTTP_X_FORWARDED_FOR']);
        $ip = trim((string)($parts[0] ?? ''));
        if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
            return $ip;
        }
    }

    $ip = $_SERVER['REMOTE_ADDR'] ?? '';
    if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
        return (string)$ip;
```

```
    }

    return null;
}

function clampStr(?string $s, int $max): string {
    $s = (string)($s ?? '');
    $s = trim($s);
    if ($s === '') {
        return '';
    }
    if (mb_strlen($s, 'UTF-8') > $max) {
        $s = mb_substr($s, 0, $max, 'UTF-8');
    }
    return $s;
}

try {
    $sid = isset($_GET['sid']) ? (int)$_GET['sid'] : 0;
    $p   = isset($_GET['p']) ? (string)$_GET['p'] : '';
    $ts  = isset($_GET['ts']) ? (string)$_GET['ts'] : '';
    $t   = isset($_GET['t']) ? (string)$_GET['t'] : '';

    if ($sid <= 0 || $p === '' || $ts === '' || $t === '') {
        outGif();
    }

    // Normalize path
    if ($p[0] !== '/') {
        $p = '/' . $p;
    }
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
    $p = clampStr($p, 1024);

    // ts sanity
    if (!ctype_digit($ts)) {
        outGif();
    }
    $tsInt = (int)$ts;
    $now = time();
```

```
// Anti-replay window (±2h)
if ($tsInt < ($now - 7200) || $tsInt > ($now + 7200)) {
    outGif();
}

// Load site secret
$stmt = $pdo->prepare('SELECT secret FROM sites WHERE id = :sid LIMIT 1');
$stmt->execute([':sid' => $sid]);
$secret = $stmt->fetchColumn();
if ($secret === false) {
    outGif();
}
$secretBin = (string)$secret; // BINARY(32)

// Verify HMAC (sid\npath\nts)
$msg = $sid . "\n" . $p . "\n" . $tsInt;
$exp = hash_hmac('sha256', $msg, $secretBin, false);
if (!hash_equals($exp, $t)) {
    outGif();
}

// Visitor identity (unique per day + page)
$ipStr = getClientIp();
$ua    = clampStr($_SERVER['HTTP_USER_AGENT'] ?? '', 512);

$vhInput = ($ipStr ?? '') . "\n" . $ua;
$visitorHash = hash('sha256', $vhInput, true); // BINARY(32)

// daily_site: hits always
$pdo->exec(
    'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
    . 'VALUES (' . (int)$sid . ', CURDATE(), 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);

// daily_page: hits always
$stmt = $pdo->prepare(
    'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
    . 'VALUES (:sid, CURDATE(), :path, 1, 0)\n'
```

```
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);
$stmt->execute([':sid' => $sid, ':path' => $p]);

// IP packed (optional)
$ipBin = null;
if ($ipStr !== null) {
    $packed = @inet_pton($ipStr);
    if ($packed !== false) {
        $ipBin = $packed;
    }
}

// unique_seen: one row per (visitor_hash + path) per day
$scope = 'page';

$stmt = $pdo->prepare(
    'INSERT INTO unique_seen\n'
    . '  (site_id, day, scope, path, visitor_hash, last_seen_at, ip, ua, hits)\n'
    . 'VALUES\n'
    . '  (:sid, CURDATE(), :scope, :path, :vh, FROM_UNIXTIME(:ts), :ip, :ua, 1)\n'
    . 'ON DUPLICATE KEY UPDATE\n'
    . '  last_seen_at = VALUES(last_seen_at),\n'
    . '  ip = VALUES(ip),\n'
    . '  ua = VALUES(ua),\n'
    . '  hits = hits + 1'
);
$stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
$stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
$stmt->bindValue(':path', $p, PDO::PARAM_STR);
$stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
$stmt->bindValue(':ts', $tsInt, PDO::PARAM_INT);
$stmt->bindValue(':ip', $ipBin, $ipBin === null ? PDO::PARAM_NULL : PDO::PARAM_LOB);
$stmt->bindValue(':ua', $ua, PDO::PARAM_STR);
$stmt->execute();

// Detect if it was a new unique for that page today (hits == 1)
$stmt = $pdo->prepare(
    'SELECT hits FROM unique_seen\n'
    . 'WHERE site_id=:sid AND day=CURDATE() AND scope=:scope AND visitor_hash=:vh AND
```

```
path=:path\n'
        . 'LIMIT 1'
    );
    $stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
    $stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
    $stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
    $stmt->bindValue(':path', $p, PDO::PARAM_STR);
    $stmt->execute();
    $h = (int)$stmt->fetchColumn();

    if ($h === 1) {
        // daily_site uniques +1
        $pdo->exec(
            'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
            . 'VALUES (' . (int)$sid . ', CURDATE(), 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );

        // daily_page uniques +1
        $stmt = $pdo->prepare(
            'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
            . 'VALUES (:sid, CURDATE(), :path, 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );
        $stmt->execute([':sid' => $sid, ':path' => $p]);
    }

    outGif();

} catch (Throwable $e) {
    error_log('[metriks i.php] ' . $e->getMessage());
    outGif();
}
PHP

# Deploy dashboard.php
cat > /var/www/metriks/public/dashboard.php <<'PHP'
<?php
declare(strict_types=1);
```

```
$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function h(string $s): string { return htmlspecialchars($s, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-
8'); }
function fmtIp(?string $bin): string {
    if ($bin === null || $bin === '') return '';
    $ip = @inet_ntop($bin);
    return $ip === false ? '' : $ip;
}

$sites = $pdo->query('SELECT id, host, label FROM sites ORDER BY id')-
>fetchAll(PDO::FETCH_ASSOC);

$siteId = isset($_GET['site']) ? (int)$_GET['site'] : (int)($sites[0]['id'] ?? 1);
$range  = (string)($_GET['range'] ?? '7d'); // today|7d|30d
if (!in_array($range, ['today', '7d', '30d'], true)) {
    $range = '7d';
}

$fromExpr = match ($range) {
    'today' => 'CURDATE()',
    '7d'    => 'DATE_SUB(CURDATE(), INTERVAL 6 DAY)',
    '30d'   => 'DATE_SUB(CURDATE(), INTERVAL 29 DAY)',
};

$siteLabel = 'site';
foreach ($sites as $s) {
    if ((int)$s['id'] === $siteId) {
        $siteLabel = (string)($s['label'] ?: $s['host']);
        break;
    }
}

// Daily series
$stmt = $pdo->prepare(
    'SELECT ds.day, ds.hits, ds.uniques\n'
```

```
    . 'FROM daily_site ds\n'
    . 'WHERE ds.site_id = :sid AND ds.day >= ' . $fromExpr . '\n'
    . 'ORDER BY ds.day DESC'
);
$stmt->execute([':sid' => $siteId]);
$series = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Top pages
$stmt = $pdo->prepare(
    'SELECT dp.path, SUM(dp.hits) AS hits, SUM(dp.uniques) AS uniques\n'
    . 'FROM daily_page dp\n'
    . 'WHERE dp.site_id = :sid AND dp.day >= ' . $fromExpr . '\n'
    . 'GROUP BY dp.path\n'
    . 'ORDER BY hits DESC\n'
    . 'LIMIT 50'
);
$stmt->execute([':sid' => $siteId]);
$topPages = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Latest visits (based on unique_seen.last_seen_at)
$latestVisits = [];
try {
    $stmt = $pdo->prepare(
        'SELECT path, last_seen_at, ip, ua, hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'ORDER BY last_seen_at DESC\n'
        . 'LIMIT 50'
    );
    $stmt->execute([':sid' => $siteId]);
    $latestVisits = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $latestVisits = [];
}

// User agents (top UA)
$userAgents = [];
try {
    $stmt = $pdo->prepare(
        'SELECT ua, SUM(hits) AS hits\n'
```

```
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'GROUP BY ua\n'
        . 'ORDER BY hits DESC\n'
        . 'LIMIT 20'
    );
    $stmt->execute([':sid' => $siteId]);
    $userAgents = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $userAgents = [];
}

$sumHits = 0;
$sumUniq = 0;
foreach ($series as $r) {
    $sumHits += (int)$r['hits'];
    $sumUniq += (int)$r['uniques'];
}

// Endpoint detection (for footer)
$endpoint = '/i.gif';
$handler = '';
try {
    $ch = curl_init('https://' . ($_SERVER['HTTP_HOST'] ?? 'localhost') . '/i.gif?x=' .
time());
    curl_setopt_array($ch, [
        CURLOPT_NOBODY => true,
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_TIMEOUT => 2,
        CURLOPT_SSL_VERIFYPEER => false,
        CURLOPT_SSL_VERIFYHOST => false,
    ]);
    curl_exec($ch);
    $handler = (string)curl_getinfo($ch, CURLINFO_HEADER_OUT);
    curl_close($ch);
} catch (Throwable) {
}
?>
<!doctype html>
<html lang="en">
```

```
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Metriks — <?=h($siteLabel)?></title>
  <link rel="stylesheet" href="/assets/css/kuzai-theme.css">
</head>
<body>

  <div class="kz-wrap">

    <header class="kz-header card">
      <div class="kz-header-title">METRIKS</div>
      <div class="kz-header-sub">Dashboard — <?=h($siteLabel)?></div>
    </header>

    <section class="card kz-toolbar">
      <form class="kz-toolbar-form" method="get" action="/dashboard.php">
        <label class="muted">Site</label>
        <select name="site">
          <?php foreach ($sites as $s): ?>
            <option value="<?= (int)$s['id'] ?>" <?= ((int)$s['id'] === $siteId ? 'selected' :
'') ?>>
              <?=h((string)($s['label'] ?: $s['host']))?>
            </option>
          <?php endforeach; ?>
        </select>

        <label class="muted">Range</label>
        <select name="range">
          <option value="today" <?=($range===='today'?'selected':'')?>>Today</option>
          <option value="7d" <?=($range===='7d'?'selected':'')?>>7 days</option>
          <option value="30d" <?=($range===='30d'?'selected':'')?>>30 days</option>
        </select>

        <button class="btn" type="submit">Show</button>
      </form>
    </section>

    <section class="kz-kpis">
      <div class="card kpi">
```

```
      <div class="muted">Hits (range)</div>
      <div class="kpi-value"><?= (int)$sumHits ?></div>
    </div>
    <div class="card kpi">
      <div class="muted">Uniques (range)</div>
      <div class="kpi-value"><?= (int)$sumUniq ?></div>
    </div>
    <div class="card kpi">
      <div class="muted">Days</div>
      <div class="kpi-value"><?= count($series) ?></div>
    </div>
  </section>

  <section class="kz-grid">
    <div class="card">
      <div class="card-title">Daily series</div>
      <table>
        <thead>
          <tr><th>Day</th><th>Hits</th><th>Uniques</th></tr>
        </thead>
        <tbody>
          <?php foreach ($series as $r): ?>
            <tr>
              <td><?=h((string)$r['day'])?></td>
              <td><?= (int)$r['hits'] ?></td>
              <td><?= (int)$r['uniques'] ?></td>
            </tr>
          <?php endforeach; ?>
          <?php if (!$series): ?>
            <tr><td colspan="3" class="muted">No data</td></tr>
          <?php endif; ?>
        </tbody>
      </table>
    </div>

    <div class="card">
      <div class="card-title">Top pages</div>
      <table>
        <thead>
          <tr><th>Path</th><th>Hits</th><th>Uniques</th></tr>
```

```
    </thead>
    <tbody>
      <?php foreach ($topPages as $r): ?>
        <tr>
          <td><?=h((string)$r['path'])?></td>
          <td><?= (int)$r['hits'] ?></td>
          <td><?= (int)$r['uniques'] ?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$topPages): ?>
        <tr><td colspan="3" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
  <div class="card-title">Latest visits</div>
  <table>
    <thead>
      <tr><th>Time</th><th>Path</th><th>IP</th><th>Hits</th><th>User-Agent</th></tr>
    </thead>
    <tbody>
      <?php foreach ($latestVisits as $r): ?>
        <tr>
          <td><?=h((string)$r['last_seen_at'])?></td>
          <td><?=h((string)$r['path'])?></td>
          <td><?=h(fmtIp($r['ip'] ?? null))?></td>
          <td><?= (int)($r['hits'] ?? 0) ?></td>
          <td class="ua"><?=h((string)($r['ua'] ?? ''))?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$latestVisits): ?>
        <tr><td colspan="5" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
```

```
        <div class="card-title">User agents</div>
        <table>
          <thead>
            <tr><th>User-Agent</th><th>Hits</th></tr>
          </thead>
          <tbody>
            <?php foreach ($userAgents as $r): ?>
              <tr>
                <td class="ua"><?=h((string)$r['ua'])?></td>
                <td><?= (int)$r['hits'] ?></td>
              </tr>
            <?php endforeach; ?>
            <?php if (!$userAgents): ?>
              <tr><td colspan="2" class="muted">No data</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>

    </section>

    <footer class="kz-footer muted">
      Endpoint: <code>/i.gif</code> → <code>/i.php</code> • DB: <code>metrics</code>
    </footer>

  </div>

</body>
</html>
PHP

# Deploy CSS theme
cat > /var/www/metriks/public/assets/css/kuzai-theme.css <<'CSS'
/* ========================
   KUZAI / KUZFOLLOW THEME
   neon white + blue
   ======================== */

:root{
  --bg0:#050914;
```

```
  --panel: rgba(10, 22, 44, 0.68);
  --panel2: rgba(10, 22, 44, 0.55);

  --text:#E8F6FF;
  --muted:#B9D7EA;

  --neon:#00D9FF;
  --neon2:#7BE9FF;
  --border: rgba(0, 217, 255, 0.55);
  --borderSoft: rgba(0, 217, 255, 0.28);

  --glow: 0 0 10px rgba(0,217,255,.45),
          0 0 22px rgba(0,217,255,.22);
  --glowStrong: 0 0 14px rgba(0,217,255,.60),
                0 0 34px rgba(0,217,255,.28);

  --radius: 16px;
  --radius2: 22px;

  --pad: 18px;
  --gap: 16px;
}

*{ box-sizing:border-box; }
html,body{ height:100%; }

body.kuzai-metriks{
  margin:0;
  color:var(--text);
  font-family: Inter, system-ui, -apple-system, Segoe UI, Roboto, Arial, sans-serif;

  background-color: var(--bg0);
  background-image: url("/assets/img/FOND-GITHUB-SEARCH2.png");
  background-size: cover;
  background-position: center;
  background-attachment: fixed;
  background-repeat:no-repeat;

  min-height:100vh;
}
```

```
body.kuzai-metriks::before{
  content:"";
  position:fixed;
  inset:0;
  background:
    radial-gradient(1200px 600px at 12% 12%, rgba(0,217,255,.18), transparent 55%),
    radial-gradient(1000px 700px at 85% 25%, rgba(123,233,255,.12), transparent 60%),
    linear-gradient(180deg, rgba(3,6,14,.58), rgba(3,6,14,.78));
  z-index:-2;
}

body.kuzai-metriks::after{
  content:"";
  position:fixed;
  inset:-10%;
  background:
    radial-gradient(900px 420px at 50% 10%, rgba(0,217,255,.10), transparent 60%),
    radial-gradient(1100px 650px at 50% 95%, rgba(0,217,255,.06), transparent 68%);
  filter: blur(2px);
  z-index:-3;
}

a{ color: var(--neon); text-decoration:none; }
a:hover{ text-shadow: var(--glow); }

.kuz-container{
  max-width: 1320px;
  margin: 0 auto;
  padding: 18px 18px 28px;
}

.row{
  display:flex;
  gap: var(--gap);
  flex-wrap: wrap;
  align-items: center;
}

.card{
```

```
  background: var(--panel);
  border: 1px solid var(--borderSoft);
  border-radius: var(--radius);
  padding: var(--pad);
  box-shadow: var(--glow);
  backdrop-filter: blur(8px);
}

.card--thin{ padding: 12px 14px; }

.title{
  font-weight:900;
  letter-spacing:.08em;
  text-transform: uppercase;
  color: var(--text);
  text-shadow: var(--glow);
}

.subtitle, .muted{
  color: var(--muted);
  opacity: .92;
}

/* ========================
   KUZFOLLOW-like top bars
   ======================== */

.kuz-topbar{
  margin: 18px 0 14px;
  padding: 16px 20px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.72), rgba(10,22,44,.58));
  border: 1px solid var(--border);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.kuz-topbar .t1{
  font-size: 30px;
  letter-spacing: .20em;
```

```
  text-transform: uppercase;
  font-weight: 900;
  text-align:center;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

.kuz-topbar .t2{
  margin-top: 6px;
  font-size: 12px;
  letter-spacing: .26em;
  text-transform: uppercase;
  text-align:center;
  color: var(--muted);
}

.kuz-subbar{
  margin: 0 0 16px;
  padding: 14px 16px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.66), rgba(10,22,44,.52));
  border: 1px solid var(--borderSoft);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.pills{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
}

.pill{
  display:inline-flex;
  align-items:center;
  gap: 8px;
  padding: 9px 12px;
  border-radius: 999px;
  border: 1px solid rgba(0,217,255,.40);
```

```
  background: rgba(0,217,255,.08);
  box-shadow: 0 0 10px rgba(0,217,255,.12);
  font-weight: 800;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  color: var(--text);
}

.pill .k{ color: var(--muted); opacity:.95; font-weight:800; }
.pill .v{ color: var(--text); font-weight:900; text-shadow: var(--glow); }

form.kuz-filters{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
  width:100%;
  margin-top: 12px;
}

/* ========================
   Controls
   ======================== */

select, input[type="text"]{
  background: rgba(4, 10, 20, .55);
  color: var(--text);
  border: 1px solid var(--borderSoft);
  border-radius: 12px;
  padding: 10px 12px;
  outline: none;
}

select:focus, input:focus{
  border-color: var(--border);
  box-shadow: var(--glow);
}

.btn{
```

```
  background: rgba(0,217,255,.10);
  color: var(--text);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 10px 14px;
  font-weight: 900;
  letter-spacing: .10em;
  text-transform: uppercase;
  cursor: pointer;
  transition: transform .12s ease, box-shadow .12s ease, background .12s ease;
}

.btn:hover{
  background: rgba(0,217,255,.18);
  box-shadow: var(--glowStrong);
  transform: translateY(-1px);
}

.btn:active{ transform: translateY(0); }

/* ========================
   KPI
   ======================== */

.kpi{
  min-width: 240px;
  flex: 0 1 auto;
}

.kpi .k{
  color: var(--muted);
  font-weight:800;
  letter-spacing:.04em;
}

.kpi .v{
  margin-top: 6px;
  font-size: 34px;
  font-weight: 900;
  color: var(--text);
```

```
  text-shadow: var(--glowStrong);
}

/* ========================
   Tables
   ========================= */

.table-wrap{ overflow:auto; }

table{
  width:100%;
  border-collapse: collapse;
  border-radius: 14px;
  overflow:hidden;
}

thead th{
  text-align:left;
  padding: 12px 10px;
  color: var(--neon2);
  font-weight: 900;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  background: rgba(0,217,255,.08);
  border-bottom: 1px solid var(--borderSoft);
}

tbody td{
  padding: 12px 10px;
  border-bottom: 1px solid rgba(0,217,255,.14);
  vertical-align: top;
}

tbody tr:hover td{
  background: rgba(0,217,255,.05);
}

code{
  background: rgba(4, 10, 20, .60);
```

```
  border: 1px solid rgba(0,217,255,.22);
  border-radius: 10px;
  padding: 4px 8px;
  color: var(--text);
}

/* grids */
.grid-2{
  display:grid;
  grid-template-columns: 1fr 1fr;
  gap: var(--gap);
}

.grid-3{
  display:grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: var(--gap);
}

@media (max-width: 1100px){
  .grid-3{ grid-template-columns: 1fr; }
}

@media (max-width: 980px){
  .grid-2{ grid-template-columns: 1fr; }
  .kpi{ min-width: 180px; }
}

.footer{
  margin-top: 16px;
  opacity:.86;
}
CSS

# Validate syntax
php -l /var/www/metriks/public/i.php
php -l /var/www/metriks/public/dashboard.php

# Reload Apache
apache2ctl configtest
```

```
systemctl reload apache2
```

8.1) /var/www/metriks/public/i.php

Below are the **current files** (collector + dashboard + CSS) to deploy.

8.1) /var/www/metriks/public/i.php

```
<?php
declare(strict_types=1);

/*
  i.php — image endpoint (tracking)
  - Validates HMAC token (sid\npath\nts)
  - Updates daily_site / daily_page / unique_seen
  - Referrer tracking: DISABLED
  - Outputs a 1x1 GIF
*/

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function outGif(): void {
    header('X-Metriks-Handler: php');
    header('Cache-Control: no-store, no-cache, must-revalidate, max-age=0');
    header('Pragma: no-cache');
    header('Content-Type: image/gif');

    // 1x1 transparent gif
    echo base64_decode('R0lGODlhAQABAPAAAAAAAAAAACH5BAEAAAAALAAAAAABAAEAAAICRAEAOw==');
    exit;
}

function getClientIp(): ?string {
    // Prefer first IP from X-Forwarded-For if present (proxy)
    if (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
        $parts = explode(',', (string)$_SERVER['HTTP_X_FORWARDED_FOR']);
        $ip = trim((string)($parts[0] ?? ''));
```

```
        if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
            return $ip;
        }
    }

    $ip = $_SERVER['REMOTE_ADDR'] ?? '';
    if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
        return (string)$ip;
    }

    return null;
}

function clampStr(?string $s, int $max): string {
    $s = (string)($s ?? '');
    $s = trim($s);
    if ($s === '') {
        return '';
    }
    if (mb_strlen($s, 'UTF-8') > $max) {
        $s = mb_substr($s, 0, $max, 'UTF-8');
    }
    return $s;
}

try {
    $sid = isset($_GET['sid']) ? (int)$_GET['sid'] : 0;
    $p   = isset($_GET['p']) ? (string)$_GET['p'] : '';
    $ts  = isset($_GET['ts']) ? (string)$_GET['ts'] : '';
    $t   = isset($_GET['t']) ? (string)$_GET['t'] : '';

    if ($sid <= 0 || $p === '' || $ts === '' || $t === '') {
        outGif();
    }

    // Normalize path
    if ($p[0] !== '/') {
        $p = '/' . $p;
    }
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
```

```
$p = clampStr($p, 1024);

// ts sanity
if (!ctype_digit($ts)) {
    outGif();
}
$tsInt = (int)$ts;
$now = time();

// Anti-replay window (±2h)
if ($tsInt < ($now - 7200) || $tsInt > ($now + 7200)) {
    outGif();
}

// Load site secret
$stmt = $pdo->prepare('SELECT secret FROM sites WHERE id = :sid LIMIT 1');
$stmt->execute([':sid' => $sid]);
$secret = $stmt->fetchColumn();
if ($secret === false) {
    outGif();
}
$secretBin = (string)$secret; // BINARY(32)

// Verify HMAC (sid\npath\nts)
$msg = $sid . "\n" . $p . "\n" . $tsInt;
$exp = hash_hmac('sha256', $msg, $secretBin, false);
if (!hash_equals($exp, $t)) {
    outGif();
}

// Visitor identity (unique per day + page)
$ipStr = getClientIp();
$ua    = clampStr($_SERVER['HTTP_USER_AGENT'] ?? '', 512);

$vhInput = ($ipStr ?? '') . "\n" . $ua;
$visitorHash = hash('sha256', $vhInput, true); // BINARY(32)

// daily_site: hits always
$pdo->exec(
    'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
```

```
    . 'VALUES (' . (int)$sid . ', CURDATE(), 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);

// daily_page: hits always
$stmt = $pdo->prepare(
    'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
    . 'VALUES (:sid, CURDATE(), :path, 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);
$stmt->execute([':sid' => $sid, ':path' => $p]);

// IP packed (optional)
$ipBin = null;
if ($ipStr !== null) {
    $packed = @inet_pton($ipStr);
    if ($packed !== false) {
        $ipBin = $packed;
    }
}

// unique_seen: one row per (visitor_hash + path) per day
$scope = 'page';

$stmt = $pdo->prepare(
    'INSERT INTO unique_seen\n'
    . '  (site_id, day, scope, path, visitor_hash, last_seen_at, ip, ua, hits)\n'
    . 'VALUES\n'
    . '  (:sid, CURDATE(), :scope, :path, :vh, FROM_UNIXTIME(:ts), :ip, :ua, 1)\n'
    . 'ON DUPLICATE KEY UPDATE\n'
    . '  last_seen_at = VALUES(last_seen_at),\n'
    . '  ip = VALUES(ip),\n'
    . '  ua = VALUES(ua),\n'
    . '  hits = hits + 1'
);
$stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
$stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
$stmt->bindValue(':path', $p, PDO::PARAM_STR);
$stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
$stmt->bindValue(':ts', $tsInt, PDO::PARAM_INT);
```

```
    $stmt->bindValue(':ip', $ipBin, $ipBin === null ? PDO::PARAM_NULL : PDO::PARAM_LOB);
    $stmt->bindValue(':ua', $ua, PDO::PARAM_STR);
    $stmt->execute();

    // Detect if it was a new unique for that page today (hits == 1)
    $stmt = $pdo->prepare(
        'SELECT hits FROM unique_seen\n'
        . 'WHERE site_id=:sid AND day=CURDATE() AND scope=:scope AND visitor_hash=:vh AND
path=:path\n'
        . 'LIMIT 1'
    );
    $stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
    $stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
    $stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
    $stmt->bindValue(':path', $p, PDO::PARAM_STR);
    $stmt->execute();
    $h = (int)$stmt->fetchColumn();

    if ($h === 1) {
        // daily_site uniques +1
        $pdo->exec(
            'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
            . 'VALUES (' . (int)$sid . ', CURDATE(), 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );

        // daily_page uniques +1
        $stmt = $pdo->prepare(
            'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
            . 'VALUES (:sid, CURDATE(), :path, 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );
        $stmt->execute([':sid' => $sid, ':path' => $p]);
    }

    outGif();

} catch (Throwable $e) {
    error_log('[metriks i.php] ' . $e->getMessage());
    outGif();
```

```
}
```

8.2) /var/www/metriks/public/dashboard.php

```
<?php
declare(strict_types=1);

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function h(string $s): string { return htmlspecialchars($s, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-
8'); }
function fmtIp(?string $bin): string {
    if ($bin === null || $bin === '') return '';
    $ip = @inet_ntop($bin);
    return $ip === false ? '' : $ip;
}

$sites = $pdo->query('SELECT id, host, label FROM sites ORDER BY id')-
>fetchAll(PDO::FETCH_ASSOC);

$siteId = isset($_GET['site']) ? (int)$_GET['site'] : (int)($sites[0]['id'] ?? 1);
$range  = (string)($_GET['range'] ?? '7d'); // today|7d|30d
if (!in_array($range, ['today', '7d', '30d'], true)) {
    $range = '7d';
}

$fromExpr = match ($range) {
    'today' => 'CURDATE()',
    '7d'    => 'DATE_SUB(CURDATE(), INTERVAL 6 DAY)',
    '30d'   => 'DATE_SUB(CURDATE(), INTERVAL 29 DAY)',
};

$siteLabel = 'site';
foreach ($sites as $s) {
    if ((int)$s['id'] === $siteId) {
        $siteLabel = (string)($s['label'] ?: $s['host']);
```

```
        break;
    }
}

// Daily series
$stmt = $pdo->prepare(
    'SELECT ds.day, ds.hits, ds.uniques\n'
    . 'FROM daily_site ds\n'
    . 'WHERE ds.site_id = :sid AND ds.day >= ' . $fromExpr . '\n'
    . 'ORDER BY ds.day DESC'
);
$stmt->execute([':sid' => $siteId]);
$series = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Top pages
$stmt = $pdo->prepare(
    'SELECT dp.path, SUM(dp.hits) AS hits, SUM(dp.uniques) AS uniques\n'
    . 'FROM daily_page dp\n'
    . 'WHERE dp.site_id = :sid AND dp.day >= ' . $fromExpr . '\n'
    . 'GROUP BY dp.path\n'
    . 'ORDER BY hits DESC\n'
    . 'LIMIT 50'
);
$stmt->execute([':sid' => $siteId]);
$topPages = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Latest visits (based on unique_seen.last_seen_at)
$latestVisits = [];
try {
    $stmt = $pdo->prepare(
        'SELECT path, last_seen_at, ip, ua, hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'ORDER BY last_seen_at DESC\n'
        . 'LIMIT 50'
    );
    $stmt->execute([':sid' => $siteId]);
    $latestVisits = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $latestVisits = [];
```

```
}

// User agents (top UA)
$userAgents = [];
try {
    $stmt = $pdo->prepare(
        'SELECT ua, SUM(hits) AS hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'GROUP BY ua\n'
        . 'ORDER BY hits DESC\n'
        . 'LIMIT 20'
    );
    $stmt->execute([':sid' => $siteId]);
    $userAgents = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $userAgents = [];
}

$sumHits = 0;
$sumUniq = 0;
foreach ($series as $r) {
    $sumHits += (int)$r['hits'];
    $sumUniq += (int)$r['uniques'];
}

// Endpoint detection (for footer)
$endpoint = '/i.gif';
$handler = '';
try {
    $ch = curl_init('https://' . ($_SERVER['HTTP_HOST'] ?? 'localhost') . '/i.gif?x=' .
time());
    curl_setopt_array($ch, [
        CURLOPT_NOBODY => true,
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_TIMEOUT => 2,
        CURLOPT_SSL_VERIFYPEER => false,
        CURLOPT_SSL_VERIFYHOST => false,
    ]);
    curl_exec($ch);
```

```
    $handler = (string)curl_getinfo($ch, CURLINFO_HEADER_OUT);
    curl_close($ch);
} catch (Throwable) {
}
?>
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Metriks — <?=h($siteLabel)?></title>
  <link rel="stylesheet" href="/assets/css/kuzai-theme.css">
</head>
<body>

  <div class="kz-wrap">

    <header class="kz-header card">
      <div class="kz-header-title">METRIKS</div>
      <div class="kz-header-sub">Dashboard — <?=h($siteLabel)?></div>
    </header>

    <section class="card kz-toolbar">
      <form class="kz-toolbar-form" method="get" action="/dashboard.php">
        <label class="muted">Site</label>
        <select name="site">
          <?php foreach ($sites as $s): ?>
            <option value="<?= (int)$s['id'] ?>" <?= ((int)$s['id'] === $siteId ? 'selected' :
'') ?>>
              <?=h((string)($s['label'] ?: $s['host']))?>
            </option>
          <?php endforeach; ?>
        </select>

        <label class="muted">Range</label>
        <select name="range">
          <option value="today" <?=($range===='today'?'selected':'')?>>Today</option>
          <option value="7d" <?=($range===='7d'?'selected':'')?>>7 days</option>
          <option value="30d" <?=($range===='30d'?'selected':'')?>>30 days</option>
        </select>
```

```
    <button class="btn" type="submit">Show</button>
  </form>
</section>

<section class="kz-kpis">
  <div class="card kpi">
    <div class="muted">Hits (range)</div>
    <div class="kpi-value"><?= (int)$sumHits ?></div>
  </div>
  <div class="card kpi">
    <div class="muted">Uniques (range)</div>
    <div class="kpi-value"><?= (int)$sumUniq ?></div>
  </div>
  <div class="card kpi">
    <div class="muted">Days</div>
    <div class="kpi-value"><?= count($series) ?></div>
  </div>
</section>

<section class="kz-grid">
  <div class="card">
    <div class="card-title">Daily series</div>
    <table>
      <thead>
        <tr><th>Day</th><th>Hits</th><th>Uniques</th></tr>
      </thead>
      <tbody>
        <?php foreach ($series as $r): ?>
          <tr>
            <td><?=h((string)$r['day'])?></td>
            <td><?= (int)$r['hits'] ?></td>
            <td><?= (int)$r['uniques'] ?></td>
          </tr>
        <?php endforeach; ?>
        <?php if (!$series): ?>
          <tr><td colspan="3" class="muted">No data</td></tr>
        <?php endif; ?>
      </tbody>
    </table>
```

```
</div>

<div class="card">
  <div class="card-title">Top pages</div>
  <table>
    <thead>
      <tr><th>Path</th><th>Hits</th><th>Uniques</th></tr>
    </thead>
    <tbody>
      <?php foreach ($topPages as $r): ?>
        <tr>
          <td><?=h((string)$r['path'])?></td>
          <td><?= (int)$r['hits'] ?></td>
          <td><?= (int)$r['uniques'] ?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$topPages): ?>
        <tr><td colspan="3" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
  <div class="card-title">Latest visits</div>
  <table>
    <thead>
      <tr><th>Time</th><th>Path</th><th>IP</th><th>Hits</th><th>User-Agent</th></tr>
    </thead>
    <tbody>
      <?php foreach ($latestVisits as $r): ?>
        <tr>
          <td><?=h((string)$r['last_seen_at'])?></td>
          <td><?=h((string)$r['path'])?></td>
          <td><?=h(fmtIp($r['ip'] ?? null))?></td>
          <td><?= (int)($r['hits'] ?? 0) ?></td>
          <td class="ua"><?=h((string)($r['ua'] ?? ''))?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$latestVisits): ?>
```

```
              <tr><td colspan="5" class="muted">No data</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>

      <div class="card">
        <div class="card-title">User agents</div>
        <table>
          <thead>
            <tr><th>User-Agent</th><th>Hits</th></tr>
          </thead>
          <tbody>
            <?php foreach ($userAgents as $r): ?>
              <tr>
                <td class="ua"><?=h((string)$r['ua'])?></td>
                <td><?= (int)$r['hits'] ?></td>
              </tr>
            <?php endforeach; ?>
            <?php if (!$userAgents): ?>
              <tr><td colspan="2" class="muted">No data</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>

    </section>

    <footer class="kz-footer muted">
      Endpoint: <code>/i.gif</code> → <code>/i.php</code> • DB: <code>metrics</code>
    </footer>

  </div>

</body>
</html>
```

8.3) /var/www/metriks/public/assets/css/kuzai-theme.css

```
/* ========================
   KUZAI / KUZFOLLOW THEME
   neon white + blue
   ========================= */

:root{
  --bg0:#050914;
  --panel: rgba(10, 22, 44, 0.68);
  --panel2: rgba(10, 22, 44, 0.55);

  --text:#E8F6FF;
  --muted:#B9D7EA;

  --neon:#00D9FF;
  --neon2:#7BE9FF;
  --border: rgba(0, 217, 255, 0.55);
  --borderSoft: rgba(0, 217, 255, 0.28);

  --glow: 0 0 10px rgba(0,217,255,.45),
          0 0 22px rgba(0,217,255,.22);
  --glowStrong: 0 0 14px rgba(0,217,255,.60),
                0 0 34px rgba(0,217,255,.28);

  --radius: 16px;
  --radius2: 22px;

  --pad: 18px;
  --gap: 16px;
}

*{ box-sizing:border-box; }
html,body{ height:100%; }

body.kuzai-metriks{
  margin:0;
  color:var(--text);
  font-family: Inter, system-ui, -apple-system, Segoe UI, Roboto, Arial, sans-serif;

  background-color: var(--bg0);
  background-image: url("/assets/img/FOND-GITHUB-SEARCH2.png");
```

```
  background-size: cover;
  background-position: center;
  background-attachment: fixed;
  background-repeat:no-repeat;

  min-height:100vh;
}

body.kuzai-metriks::before{
  content:"";
  position:fixed;
  inset:0;
  background:
    radial-gradient(1200px 600px at 12% 12%, rgba(0,217,255,.18), transparent 55%),
    radial-gradient(1000px 700px at 85% 25%, rgba(123,233,255,.12), transparent 60%),
    linear-gradient(180deg, rgba(3,6,14,.58), rgba(3,6,14,.78));
  z-index:-2;
}

body.kuzai-metriks::after{
  content:"";
  position:fixed;
  inset:-10%;
  background:
    radial-gradient(900px 420px at 50% 10%, rgba(0,217,255,.10), transparent 60%),
    radial-gradient(1100px 650px at 50% 95%, rgba(0,217,255,.06), transparent 68%);
  filter: blur(2px);
  z-index:-3;
}

a{ color: var(--neon); text-decoration:none; }
a:hover{ text-shadow: var(--glow); }

.kuz-container{
  max-width: 1320px;
  margin: 0 auto;
  padding: 18px 18px 28px;
}

.row{
```

```
  display:flex;
  gap: var(--gap);
  flex-wrap: wrap;
  align-items: center;
}

.card{
  background: var(--panel);
  border: 1px solid var(--borderSoft);
  border-radius: var(--radius);
  padding: var(--pad);
  box-shadow: var(--glow);
  backdrop-filter: blur(8px);
}

.card--thin{ padding: 12px 14px; }

.title{
  font-weight:900;
  letter-spacing:.08em;
  text-transform: uppercase;
  color: var(--text);
  text-shadow: var(--glow);
}

.subtitle, .muted{
  color: var(--muted);
  opacity: .92;
}

/* ========================
   KUZFOLLOW-like top bars
   ======================== */

.kuz-topbar{
  margin: 18px 0 14px;
  padding: 16px 20px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.72), rgba(10,22,44,.58));
  border: 1px solid var(--border);
```

```
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.kuz-topbar .t1{
  font-size: 30px;
  letter-spacing: .20em;
  text-transform: uppercase;
  font-weight: 900;
  text-align:center;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

.kuz-topbar .t2{
  margin-top: 6px;
  font-size: 12px;
  letter-spacing: .26em;
  text-transform: uppercase;
  text-align:center;
  color: var(--muted);
}

.kuz-subbar{
  margin: 0 0 16px;
  padding: 14px 16px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.66), rgba(10,22,44,.52));
  border: 1px solid var(--borderSoft);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.pills{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
}
```

```
.pill{
  display:inline-flex;
  align-items:center;
  gap: 8px;
  padding: 9px 12px;
  border-radius: 999px;
  border: 1px solid rgba(0,217,255,.40);
  background: rgba(0,217,255,.08);
  box-shadow: 0 0 10px rgba(0,217,255,.12);
  font-weight: 800;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  color: var(--text);
}

.pill .k{ color: var(--muted); opacity:.95; font-weight:800; }
.pill .v{ color: var(--text); font-weight:900; text-shadow: var(--glow); }

form.kuz-filters{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
  width:100%;
  margin-top: 12px;
}

/* ========================
   Controls
   ======================== */

select, input[type="text"]{
  background: rgba(4, 10, 20, .55);
  color: var(--text);
  border: 1px solid var(--borderSoft);
  border-radius: 12px;
  padding: 10px 12px;
  outline: none;
}
```

```
select:focus, input:focus{
  border-color: var(--border);
  box-shadow: var(--glow);
}

.btn{
  background: rgba(0,217,255,.10);
  color: var(--text);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 10px 14px;
  font-weight: 900;
  letter-spacing: .10em;
  text-transform: uppercase;
  cursor: pointer;
  transition: transform .12s ease, box-shadow .12s ease, background .12s ease;
}

.btn:hover{
  background: rgba(0,217,255,.18);
  box-shadow: var(--glowStrong);
  transform: translateY(-1px);
}

.btn:active{ transform: translateY(0); }

/* ========================
   KPI
   ======================== */

.kpi{
  min-width: 240px;
  flex: 0 1 auto;
}

.kpi .k{
  color: var(--muted);
  font-weight:800;
  letter-spacing:.04em;
```

```
}

.kpi .v{
  margin-top: 6px;
  font-size: 34px;
  font-weight: 900;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

/* ========================
   Tables
   ========================= */

.table-wrap{ overflow:auto; }

table{
  width:100%;
  border-collapse: collapse;
  border-radius: 14px;
  overflow:hidden;
}

thead th{
  text-align:left;
  padding: 12px 10px;
  color: var(--neon2);
  font-weight: 900;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  background: rgba(0,217,255,.08);
  border-bottom: 1px solid var(--borderSoft);
}

tbody td{
  padding: 12px 10px;
  border-bottom: 1px solid rgba(0,217,255,.14);
  vertical-align: top;
}
```

```
tbody tr:hover td{
  background: rgba(0,217,255,.05);
}

code{
  background: rgba(4, 10, 20, .60);
  border: 1px solid rgba(0,217,255,.22);
  border-radius: 10px;
  padding: 4px 8px;
  color: var(--text);
}

/* grids */
.grid-2{
  display:grid;
  grid-template-columns: 1fr 1fr;
  gap: var(--gap);
}

.grid-3{
  display:grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: var(--gap);
}

@media (max-width: 1100px){
  .grid-3{ grid-template-columns: 1fr; }
}

@media (max-width: 980px){
  .grid-2{ grid-template-columns: 1fr; }
  .kpi{ min-width: 180px; }
}

.footer{
  margin-top: 16px;
  opacity:.86;
}
```

9) Post-deploy verification checklist

```
# Apache config sanity
apache2ctl configtest
systemctl reload apache2

# Collector endpoint (must be PHP)
curl -I "https://metriks.kuzlab.org/i.gif?x=$(date +%s)" | head -n 40

# Dashboard
curl -I "https://metriks.kuzlab.org/dashboard.php" | head -n 25

# DB growth
mysql -N metrics -e "SELECT COUNT(*) AS daily_site_rows FROM daily_site;"
mysql -N metrics -e "SELECT COUNT(*) AS daily_page_rows FROM daily_page;"
mysql -N metrics -e "SELECT COUNT(*) AS unique_seen_rows FROM unique_seen;"
```

KUZAI.ORG - KUZ-METRIKS

OVERVIEW

This document describes the full installation and integration process for **Metriks** (tracking endpoint + MySQL storage + web dashboard) as deployed on: - `metriks.kuzlab.org` (collector + dashboard) - `kuzai.org` (SID=1) - `mob.kuzai.org` (SID=2) - `radio.kuzai.org` (SID=3)

INSTALLATION ACTION LIST

- Provision MySQL database `metrics` and a dedicated user.
- Create core tables: `sites`, `daily_site`, `daily_page`, `unique_seen`.
- Register each tracked site (SID/host/label) with a 32-byte secret.
- Deploy collector webroot to `/var/www/metriks`.
- Configure Apache vhosts for `metriks.kuzlab.org` (HTTP→HTTPS, PHP enabled).
- Route `/i.gif` to `/i.php` (PHP collector).
- Deploy `dashboard.php` and `index.php` redirect.
- Deploy KUZAI theme CSS and background image assets.
- Deploy `beacon.php` on each tracked site (kuzai/mob/radio) with correct constants.
- Inject beacon image/JS in WordPress and AzuraCast public pages.
- Verify collector headers with `curl -I` (`X-Metriks-Handler: php`).
- Verify end-to-end metrics with a signed test request and DB queries.

0) FILES AND DIRECTORIES

Project layout (collector host):

```
/var/www/metriks
├── config
│   └── config.php
├── public
│   ├── assets
│   │   ├── css
│   │   │   └── kuzai-theme.css
│   │   └── img
│   │       └── FOND-GITHUB-SEARCH2.png
│   ├── dashboard.php
│   ├── i.gif
│   ├── i.php
│   └── index.php
└── src
```

```
    ├── Auth.php
    ├── Db.php
    └── Metrics.php
```

---

1) MYSQL DATABASE AND USER

Create the DB and a dedicated user.

```
mysql -e "CREATE DATABASE IF NOT EXISTS metrics CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;"

mysql -e "CREATE USER IF NOT EXISTS 'metrics_user'@'localhost' IDENTIFIED BY '$SECRET';"
mysql -e "GRANT ALL PRIVILEGES ON metrics.* TO 'metrics_user'@'localhost';"
mysql -e "FLUSH PRIVILEGES;"
```

Verification:

```
mysql -e "SHOW DATABASES LIKE 'metrics';"
mysql -e "SHOW GRANTS FOR 'metrics_user'@'localhost';"
```

---

2) TABLES

Core tables used by the collector + dashboard.

```
mysql metrics <<'SQL'
CREATE TABLE IF NOT EXISTS sites (
  id INT NOT NULL AUTO_INCREMENT,
  host VARCHAR(255) NOT NULL,
  label VARCHAR(255) NULL,
  secret BINARY(32) NOT NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uniq_host (host)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE IF NOT EXISTS daily_site (
  site_id INT NOT NULL,
  day DATE NOT NULL,
  hits INT NOT NULL DEFAULT 0,
  uniques INT NOT NULL DEFAULT 0,
  PRIMARY KEY (site_id, day),
```

```
  CONSTRAINT fk_daily_site_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE IF NOT EXISTS daily_page (
  site_id INT NOT NULL,
  day DATE NOT NULL,
  path VARCHAR(1024) NOT NULL,
  hits INT NOT NULL DEFAULT 0,
  uniques INT NOT NULL DEFAULT 0,
  PRIMARY KEY (site_id, day, path(191)),
  CONSTRAINT fk_daily_page_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- unique_seen stores one row per visitor+page per day.
-- Columns shown here match the current collector/dashboard usage.
CREATE TABLE IF NOT EXISTS unique_seen (
  site_id INT NOT NULL,
  day DATE NOT NULL,
  scope ENUM('site','page') NOT NULL,
  path VARCHAR(1024) NOT NULL,
  visitor_hash BINARY(32) NOT NULL,
  last_seen_at DATETIME NOT NULL,
  ip VARBINARY(16) NULL,
  ua VARCHAR(512) NOT NULL,
  hits INT NOT NULL DEFAULT 1,
  PRIMARY KEY (site_id, day, scope, visitor_hash, path(191)),
  CONSTRAINT fk_unique_seen_site FOREIGN KEY (site_id) REFERENCES sites(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
SQL
```

Verification:

```
mysql -N metrics -e "SHOW TABLES;"
mysql -N metrics -e "SHOW CREATE TABLE unique_seen\G" | sed -n '1,160p'
```

---

3) SITES REGISTRATION (SIDS + SECRETS)

Insert the sites and generate a 32-byte secret per site.

Example (generate secrets):

```
# 32 RANDOM BYTES (HEX)
openssl rand -hex 32
```

Insert (example values):

```
mysql metrics <<'SQL'
INSERT INTO sites (id, host, label, secret) VALUES
  (1, 'kuzai.org',       'KUZAI',       UNHEX('$SECRET')),
  (2, 'mob.kuzai.org',   'KUZAI-MOB',   UNHEX('$SECRET')),
  (3, 'radio.kuzai.org', 'KUZAI-RADIO', UNHEX('$SECRET<REST_OF_HEX>'))
ON DUPLICATE KEY UPDATE host=VALUES(host), label=VALUES(label), secret=VALUES(secret);
SQL
```

Verification:

```
mysql -N metrics -e "SELECT id, host, label, LENGTH(secret) AS secret_len FROM sites ORDER BY id;"
```

---

4) APACHE VHOST FOR METRIKS.KUZLAB.ORG

Collector and dashboard are served from `/var/www/metriks/public`.

Key requirements: - HTTPS enabled (Let's Encrypt) - `/i.gif` must be handled by PHP (`/i.php`) so that it increments DB counters

Example check (vhosts + modules):

```
apache2ctl -S | grep -E "metriks\.kuzlab\.org|:80|:443" -n || true
apache2ctl -M | grep -E 'rewrite|headers|php|proxy_fcgi' || true
```

Verification (endpoint must show `X-Metriks-Handler: php`):

```
curl -I "https://metriks.kuzlab.org/i.php?x=$(date +%s)" | head -n 40
curl -I "https://metriks.kuzlab.org/i.gif?x=$(date +%s)" | head -n 40
```

---

5) BEACON INTEGRATION (KUZAI.ORG / MOB.KUZAI.ORG / RADIO.KUZAI.ORG)

Each site exposes a local `beacon.php` that: - computes a signed token - redirects to `https://metriks.kuzlab.org/i.gif?...`

Important: referrer is not used. The path is passed explicitly using `?p=<path>`.

5.1) BEACON.PHP (TEMPLATE)

Deploy this file on each site as `/beacon.php` and set `SID` and `SECRET_HEX`.

```php
<?php
declare(strict_types=1);

// ===== METRIKS BEACON (NO REFERRER) =====
// Per-site constants
const SID = 0; // TODO: set 1/2/3
const SECRET_HEX = '$SECRET'; // TODO: 64 hex chars
const METRICS_BASE = 'https://metriks.kuzlab.org/i.gif';

function clampPath(string $p): string {
    $p = trim($p);
    if ($p === '') return '/';
    if ($p[0] !== '/') $p = '/' . $p;
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
    if (mb_strlen($p, 'UTF-8') > 1024) $p = mb_substr($p, 0, 1024, 'UTF-8');
    return $p;
}

$path = clampPath((string)($_GET['p'] ?? '/'));
$ts   = (string)time();

$secret = hex2bin(SECRET_HEX);
if ($secret === false) {
    http_response_code(500);
    exit;
}

$msg = SID . "\n" . $path . "\n" . $ts;
$tok = hash_hmac('sha256', $msg, $secret, false);

$url = METRICS_BASE
    . '?sid=' . rawurlencode((string)SID)
    . '&p='   . rawurlencode($path)
    . '&ts='  . rawurlencode($ts)
    . '&t='   . rawurlencode($tok);

header('Location: ' . $url, true, 302);
```

```
exit;
```

5.2) SET CONSTANTS PER SITE

- `kuzai.org`:
  - `const SID = 1;`
  - `const SECRET_HEX = '$SECRET';`
- `mob.kuzai.org`:
  - `const SID = 2;`
  - `const SECRET_HEX = '$SECRET';`
- `radio.kuzai.org`:
  - `const SID = 3;`
  - `const SECRET_HEX = '$SECRET';`

5.3) FRONT-END INJECTION

WordPress (theme): insert a hidden image with the current path.

```
<img src="/beacon.php?p=/" style="display:none" alt="">
<script>
  (function(){
    var i = document.createElement('img');
    i.src = '/beacon.php?p=' + encodeURIComponent(location.pathname) + '&x=' + Date.now();
    i.alt = '';
    i.style.display = 'none';
    document.body.appendChild(i);
  })();
</script>
```

AzuraCast (Public Pages → Custom JS):

```
// === METRIKS BEACON (radio.kuzai.org) ===
(function () {
  if (document.getElementById('metriks-beacon')) return;

  const img = document.createElement('img');
  img.id = 'metriks-beacon';
  img.alt = '';
  img.width = 1;
  img.height = 1;
  img.style.position = 'absolute';
  img.style.left = '-9999px';
  img.style.width = '1px';
```

```
  img.style.height = '1px';
  img.referrerPolicy = 'strict-origin-when-cross-origin';

  const p = encodeURIComponent(location.pathname || '/');
  img.src = 'https://radio.kuzai.org/beacon.php?p=' + p + '&x=' + Date.now();

  const target = document.body || document.head || document.documentElement;
  if (target) target.appendChild(img);
})();
```

Verification (should return a 302 to metriks i.gif):

```
curl -I "https://mob.kuzai.org/beacon.php?p=%2F&x=$(date +%s)" | head -n 25
curl -I "https://radio.kuzai.org/beacon.php?p=%2F&x=$(date +%s)" | head -n 25
```

---

6) MANUAL COLLECTOR TEST (DIRECT HIT)

Generate a valid token and call the collector.

```
set -euo pipefail

SID=1
P="/manual-test"
TS="$(date +%s)"
SECRET_HEX="$(mysql -N metrics -e "SELECT HEX(secret) FROM sites WHERE id=${SID};")"

TOKEN="$(SID="$SID" P="$P" TS="$TS" SECRET_HEX="$SECRET_HEX" php -r '
$sid=(int)getenv("SID");
$p=getenv("P");
$ts=getenv("TS");
$secret=hex2bin(getenv("SECRET_HEX"));
echo hash_hmac("sha256", $sid."\n".$p."\n".$ts, $secret, false);
')"

P_ENC="$(P="$P" php -r 'echo rawurlencode(getenv("P"));')"

curl -s -o /dev/null
"https://metriks.kuzlab.org/i.gif?sid=${SID}&p=${P_ENC}&ts=${TS}&t=${TOKEN}"

mysql -N metrics -e "SELECT site_id, day, hits, uniques FROM daily_site WHERE site_id=${SID}
```

```
AND day=CURDATE();"
mysql -N metrics -e "SELECT path, hits, uniques FROM daily_page WHERE site_id=${SID} AND
day=CURDATE() ORDER BY hits DESC LIMIT 10;"
mysql -N metrics -e "SELECT COUNT(*) FROM unique_seen;"
```

---

7) WEB APP (DASHBOARD)

- `/dashboard.php`: UI (KUZAI theme)
- `/index.php`: redirect to the dashboard

Verification:

```
curl -I https://metriks.kuzlab.org/dashboard.php | head -n 25
curl -I https://metriks.kuzlab.org/ | head -n 20
```

---

8) DEPLOY/UPDATE APPLICATION FILES

Run these commands on the collector host to deploy the current code.

```
set -euo pipefail

# BACKUP CURRENT FILES
cp -a /var/www/metriks/public/i.php /var/www/metriks/public/i.php.bk.$(date +%s) 2>/dev/null
|| true
cp -a /var/www/metriks/public/dashboard.php /var/www/metriks/public/dashboard.php.bk.$(date
+%s) 2>/dev/null || true
cp -a /var/www/metriks/public/assets/css/kuzai-theme.css
/var/www/metriks/public/assets/css/kuzai-theme.css.bk.$(date +%s) 2>/dev/null || true

# DEPLOY I.PHP
cat > /var/www/metriks/public/i.php <<'PHP'
<?php
declare(strict_types=1);

/*
  i.php — image endpoint (tracking)
  - Validates HMAC token (sid\npath\nts)
  - Updates daily_site / daily_page / unique_seen
  - Referrer tracking: DISABLED
  - Outputs a 1x1 GIF
```

```
*/

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function outGif(): void {
    header('X-Metriks-Handler: php');
    header('Cache-Control: no-store, no-cache, must-revalidate, max-age=0');
    header('Pragma: no-cache');
    header('Content-Type: image/gif');

    // 1x1 transparent gif
    echo base64_decode('R0lGODlhAQABAPAAAAAAAAAAACH5BAEAAAAALAAAAAABAAEAAAICRAEAOw==');
    exit;
}

function getClientIp(): ?string {
    // Prefer first IP from X-Forwarded-For if present (proxy)
    if (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
        $parts = explode(',', (string)$_SERVER['HTTP_X_FORWARDED_FOR']);
        $ip = trim((string)($parts[0] ?? ''));
        if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
            return $ip;
        }
    }

    $ip = $_SERVER['REMOTE_ADDR'] ?? '';
    if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
        return (string)$ip;
    }

    return null;
}

function clampStr(?string $s, int $max): string {
    $s = (string)($s ?? '');
    $s = trim($s);
```

```
    if ($s === '') {
        return '';
    }
    if (mb_strlen($s, 'UTF-8') > $max) {
        $s = mb_substr($s, 0, $max, 'UTF-8');
    }
    return $s;
}

try {
    $sid = isset($_GET['sid']) ? (int)$_GET['sid'] : 0;
    $p   = isset($_GET['p']) ? (string)$_GET['p'] : '';
    $ts  = isset($_GET['ts']) ? (string)$_GET['ts'] : '';
    $t   = isset($_GET['t']) ? (string)$_GET['t'] : '';

    if ($sid <= 0 || $p === '' || $ts === '' || $t === '') {
        outGif();
    }

    // Normalize path
    if ($p[0] !== '/') {
        $p = '/' . $p;
    }
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
    $p = clampStr($p, 1024);

    // ts sanity
    if (!ctype_digit($ts)) {
        outGif();
    }
    $tsInt = (int)$ts;
    $now = time();

    // Anti-replay window (±2h)
    if ($tsInt < ($now - 7200) || $tsInt > ($now + 7200)) {
        outGif();
    }

    // Load site secret
    $stmt = $pdo->prepare('SELECT secret FROM sites WHERE id = :sid LIMIT 1');
```

```
$stmt->execute([':sid' => $sid]);
$secret = $stmt->fetchColumn();
if ($secret === false) {
    outGif();
}
$secretBin = (string)$secret; // BINARY(32)

// Verify HMAC (sid\npath\nts)
$msg = $sid . "\n" . $p . "\n" . $tsInt;
$exp = hash_hmac('sha256', $msg, $secretBin, false);
if (!hash_equals($exp, $t)) {
    outGif();
}

// Visitor identity (unique per day + page)
$ipStr = getClientIp();
$ua    = clampStr($_SERVER['HTTP_USER_AGENT'] ?? '', 512);

$vhInput = ($ipStr ?? '') . "\n" . $ua;
$visitorHash = hash('sha256', $vhInput, true); // BINARY(32)

// daily_site: hits always
$pdo->exec(
    'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
    . 'VALUES (' . (int)$sid . ', CURDATE(), 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);

// daily_page: hits always
$stmt = $pdo->prepare(
    'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
    . 'VALUES (:sid, CURDATE(), :path, 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);
$stmt->execute([':sid' => $sid, ':path' => $p]);

// IP packed (optional)
$ipBin = null;
if ($ipStr !== null) {
    $packed = @inet_pton($ipStr);
```

```
        if ($packed !== false) {
            $ipBin = $packed;
        }
    }

    // unique_seen: one row per (visitor_hash + path) per day
    $scope = 'page';

    $stmt = $pdo->prepare(
        'INSERT INTO unique_seen\n'
        . '  (site_id, day, scope, path, visitor_hash, last_seen_at, ip, ua, hits)\n'
        . 'VALUES\n'
        . '  (:sid, CURDATE(), :scope, :path, :vh, FROM_UNIXTIME(:ts), :ip, :ua, 1)\n'
        . 'ON DUPLICATE KEY UPDATE\n'
        . '  last_seen_at = VALUES(last_seen_at),\n'
        . '  ip = VALUES(ip),\n'
        . '  ua = VALUES(ua),\n'
        . '  hits = hits + 1'
    );
    $stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
    $stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
    $stmt->bindValue(':path', $p, PDO::PARAM_STR);
    $stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
    $stmt->bindValue(':ts', $tsInt, PDO::PARAM_INT);
    $stmt->bindValue(':ip', $ipBin, $ipBin === null ? PDO::PARAM_NULL : PDO::PARAM_LOB);
    $stmt->bindValue(':ua', $ua, PDO::PARAM_STR);
    $stmt->execute();

    // Detect if it was a new unique for that page today (hits == 1)
    $stmt = $pdo->prepare(
        'SELECT hits FROM unique_seen\n'
        . 'WHERE site_id=:sid AND day=CURDATE() AND scope=:scope AND visitor_hash=:vh AND
path=:path\n'
        . 'LIMIT 1'
    );
    $stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
    $stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
    $stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
    $stmt->bindValue(':path', $p, PDO::PARAM_STR);
    $stmt->execute();
```

```
    $h = (int)$stmt->fetchColumn();

    if ($h === 1) {
        // daily_site uniques +1
        $pdo->exec(
            'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
            . 'VALUES (' . (int)$sid . ', CURDATE(), 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );

        // daily_page uniques +1
        $stmt = $pdo->prepare(
            'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
            . 'VALUES (:sid, CURDATE(), :path, 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );
        $stmt->execute([':sid' => $sid, ':path' => $p]);
    }

    outGif();

} catch (Throwable $e) {
    error_log('[metriks i.php] ' . $e->getMessage());
    outGif();
}
PHP

# DEPLOY DASHBOARD.PHP
cat > /var/www/metriks/public/dashboard.php <<'PHP'
<?php
declare(strict_types=1);

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function h(string $s): string { return htmlspecialchars($s, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-8'); }
```

```
function fmtIp(?string $bin): string {
    if ($bin === null || $bin === '') return '';
    $ip = @inet_ntop($bin);
    return $ip === false ? '' : $ip;
}

$sites = $pdo->query('SELECT id, host, label FROM sites ORDER BY id')-
>fetchAll(PDO::FETCH_ASSOC);

$siteId = isset($_GET['site']) ? (int)$_GET['site'] : (int)($sites[0]['id'] ?? 1);
$range  = (string)($_GET['range'] ?? '7d'); // today|7d|30d
if (!in_array($range, ['today', '7d', '30d'], true)) {
    $range = '7d';
}

$fromExpr = match ($range) {
    'today' => 'CURDATE()',
    '7d'    => 'DATE_SUB(CURDATE(), INTERVAL 6 DAY)',
    '30d'   => 'DATE_SUB(CURDATE(), INTERVAL 29 DAY)',
};

$siteLabel = 'site';
foreach ($sites as $s) {
    if ((int)$s['id'] === $siteId) {
        $siteLabel = (string)($s['label'] ?: $s['host']);
        break;
    }
}

// Daily series
$stmt = $pdo->prepare(
    'SELECT ds.day, ds.hits, ds.uniques\n'
    . 'FROM daily_site ds\n'
    . 'WHERE ds.site_id = :sid AND ds.day >= ' . $fromExpr . '\n'
    . 'ORDER BY ds.day DESC'
);
$stmt->execute([':sid' => $siteId]);
$series = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Top pages
```

```
$stmt = $pdo->prepare(
    'SELECT dp.path, SUM(dp.hits) AS hits, SUM(dp.uniques) AS uniques\n'
    . 'FROM daily_page dp\n'
    . 'WHERE dp.site_id = :sid AND dp.day >= ' . $fromExpr . '\n'
    . 'GROUP BY dp.path\n'
    . 'ORDER BY hits DESC\n'
    . 'LIMIT 50'
);
$stmt->execute([':sid' => $siteId]);
$topPages = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Latest visits (based on unique_seen.last_seen_at)
$latestVisits = [];
try {
    $stmt = $pdo->prepare(
        'SELECT path, last_seen_at, ip, ua, hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'ORDER BY last_seen_at DESC\n'
        . 'LIMIT 50'
    );
    $stmt->execute([':sid' => $siteId]);
    $latestVisits = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $latestVisits = [];
}

// User agents (top UA)
$userAgents = [];
try {
    $stmt = $pdo->prepare(
        'SELECT ua, SUM(hits) AS hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'GROUP BY ua\n'
        . 'ORDER BY hits DESC\n'
        . 'LIMIT 20'
    );
    $stmt->execute([':sid' => $siteId]);
    $userAgents = $stmt->fetchAll(PDO::FETCH_ASSOC);
```

```
} catch (Throwable) {
    $userAgents = [];
}

$sumHits = 0;
$sumUniq = 0;
foreach ($series as $r) {
    $sumHits += (int)$r['hits'];
    $sumUniq += (int)$r['uniques'];
}

// Endpoint detection (for footer)
$endpoint = '/i.gif';
$handler = '';
try {
    $ch = curl_init('https://' . ($_SERVER['HTTP_HOST'] ?? 'localhost') . '/i.gif?x=' .
time());
    curl_setopt_array($ch, [
        CURLOPT_NOBODY => true,
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_TIMEOUT => 2,
        CURLOPT_SSL_VERIFYPEER => false,
        CURLOPT_SSL_VERIFYHOST => false,
    ]);
    curl_exec($ch);
    $handler = (string)curl_getinfo($ch, CURLINFO_HEADER_OUT);
    curl_close($ch);
} catch (Throwable) {
}
?>
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Metriks — <?=h($siteLabel)?></title>
  <link rel="stylesheet" href="/assets/css/kuzai-theme.css">
</head>
<body>
```

```
  <div class="kz-wrap">

    <header class="kz-header card">
      <div class="kz-header-title">METRIKS</div>
      <div class="kz-header-sub">Dashboard — <?=h($siteLabel)?></div>
    </header>

    <section class="card kz-toolbar">
      <form class="kz-toolbar-form" method="get" action="/dashboard.php">
        <label class="muted">Site</label>
        <select name="site">
          <?php foreach ($sites as $s): ?>
            <option value="<?= (int)$s['id'] ?>" <?= ((int)$s['id'] === $siteId ? 'selected' :
'') ?>>
              <?=h((string)($s['label'] ?: $s['host']))?>
            </option>
          <?php endforeach; ?>
        </select>

        <label class="muted">Range</label>
        <select name="range">
          <option value="today" <?=($range===='today'?'selected':'')?>>Today</option>
          <option value="7d" <?=($range===='7d'?'selected':'')?>>7 days</option>
          <option value="30d" <?=($range===='30d'?'selected':'')?>>30 days</option>
        </select>

        <button class="btn" type="submit">Show</button>
      </form>
    </section>

    <section class="kz-kpis">
      <div class="card kpi">
        <div class="muted">Hits (range)</div>
        <div class="kpi-value"><?= (int)$sumHits ?></div>
      </div>
      <div class="card kpi">
        <div class="muted">Uniques (range)</div>
        <div class="kpi-value"><?= (int)$sumUniq ?></div>
      </div>
      <div class="card kpi">
```

```
    <div class="muted">Days</div>
    <div class="kpi-value"><?= count($series) ?></div>
  </div>
</section>

<section class="kz-grid">
  <div class="card">
    <div class="card-title">Daily series</div>
    <table>
      <thead>
        <tr><th>Day</th><th>Hits</th><th>Uniques</th></tr>
      </thead>
      <tbody>
        <?php foreach ($series as $r): ?>
          <tr>
            <td><?=h((string)$r['day'])?></td>
            <td><?= (int)$r['hits'] ?></td>
            <td><?= (int)$r['uniques'] ?></td>
          </tr>
        <?php endforeach; ?>
        <?php if (!$series): ?>
          <tr><td colspan="3" class="muted">No data</td></tr>
        <?php endif; ?>
      </tbody>
    </table>
  </div>

  <div class="card">
    <div class="card-title">Top pages</div>
    <table>
      <thead>
        <tr><th>Path</th><th>Hits</th><th>Uniques</th></tr>
      </thead>
      <tbody>
        <?php foreach ($topPages as $r): ?>
          <tr>
            <td><?=h((string)$r['path'])?></td>
            <td><?= (int)$r['hits'] ?></td>
            <td><?= (int)$r['uniques'] ?></td>
          </tr>
```

```
      <?php endforeach; ?>
      <?php if (!$topPages): ?>
        <tr><td colspan="3" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
  <div class="card-title">Latest visits</div>
  <table>
    <thead>
      <tr><th>Time</th><th>Path</th><th>IP</th><th>Hits</th><th>User-Agent</th></tr>
    </thead>
    <tbody>
      <?php foreach ($latestVisits as $r): ?>
        <tr>
          <td><?=h((string)$r['last_seen_at'])?></td>
          <td><?=h((string)$r['path'])?></td>
          <td><?=h(fmtIp($r['ip'] ?? null))?></td>
          <td><?= (int)($r['hits'] ?? 0) ?></td>
          <td class="ua"><?=h((string)($r['ua'] ?? ''))?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$latestVisits): ?>
        <tr><td colspan="5" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
  <div class="card-title">User agents</div>
  <table>
    <thead>
      <tr><th>User-Agent</th><th>Hits</th></tr>
    </thead>
    <tbody>
      <?php foreach ($userAgents as $r): ?>
        <tr>
```

```
                <td class="ua"><?=h((string)$r['ua'])?></td>
                <td><?= (int)$r['hits'] ?></td>
              </tr>
            <?php endforeach; ?>
            <?php if (!$userAgents): ?>
              <tr><td colspan="2" class="muted">No data</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>

    </section>

    <footer class="kz-footer muted">
      Endpoint: <code>/i.gif</code> → <code>/i.php</code> • DB: <code>metrics</code>
    </footer>

  </div>

</body>
</html>
PHP

# DEPLOY CSS THEME
cat > /var/www/metriks/public/assets/css/kuzai-theme.css <<'CSS'
/* ========================
   KUZAI / KUZFOLLOW THEME
   neon white + blue
   ========================= */

:root{
  --bg0:#050914;
  --panel: rgba(10, 22, 44, 0.68);
  --panel2: rgba(10, 22, 44, 0.55);

  --text:#E8F6FF;
  --muted:#B9D7EA;

  --neon:#00D9FF;
  --neon2:#7BE9FF;
```

```
  --border: rgba(0, 217, 255, 0.55);
  --borderSoft: rgba(0, 217, 255, 0.28);

  --glow: 0 0 10px rgba(0,217,255,.45),
          0 0 22px rgba(0,217,255,.22);
  --glowStrong: 0 0 14px rgba(0,217,255,.60),
                0 0 34px rgba(0,217,255,.28);

  --radius: 16px;
  --radius2: 22px;

  --pad: 18px;
  --gap: 16px;
}

*{ box-sizing:border-box; }
html,body{ height:100%; }

body.kuzai-metriks{
  margin:0;
  color:var(--text);
  font-family: Inter, system-ui, -apple-system, Segoe UI, Roboto, Arial, sans-serif;

  background-color: var(--bg0);
  background-image: url("/assets/img/FOND-GITHUB-SEARCH2.png");
  background-size: cover;
  background-position: center;
  background-attachment: fixed;
  background-repeat:no-repeat;

  min-height:100vh;
}

body.kuzai-metriks::before{
  content:"";
  position:fixed;
  inset:0;
  background:
    radial-gradient(1200px 600px at 12% 12%, rgba(0,217,255,.18), transparent 55%),
    radial-gradient(1000px 700px at 85% 25%, rgba(123,233,255,.12), transparent 60%),
```

```
    linear-gradient(180deg, rgba(3,6,14,.58), rgba(3,6,14,.78));
  z-index:-2;
}

body.kuzai-metriks::after{
  content:"";
  position:fixed;
  inset:-10%;
  background:
    radial-gradient(900px 420px at 50% 10%, rgba(0,217,255,.10), transparent 60%),
    radial-gradient(1100px 650px at 50% 95%, rgba(0,217,255,.06), transparent 68%);
  filter: blur(2px);
  z-index:-3;
}

a{ color: var(--neon); text-decoration:none; }
a:hover{ text-shadow: var(--glow); }

.kuz-container{
  max-width: 1320px;
  margin: 0 auto;
  padding: 18px 18px 28px;
}

.row{
  display:flex;
  gap: var(--gap);
  flex-wrap: wrap;
  align-items: center;
}

.card{
  background: var(--panel);
  border: 1px solid var(--borderSoft);
  border-radius: var(--radius);
  padding: var(--pad);
  box-shadow: var(--glow);
  backdrop-filter: blur(8px);
}
```

```
.card--thin{ padding: 12px 14px; }

.title{
  font-weight:900;
  letter-spacing:.08em;
  text-transform: uppercase;
  color: var(--text);
  text-shadow: var(--glow);
}

.subtitle, .muted{
  color: var(--muted);
  opacity: .92;
}

/* ========================
   KUZFOLLOW-like top bars
   ======================== */

.kuz-topbar{
  margin: 18px 0 14px;
  padding: 16px 20px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.72), rgba(10,22,44,.58));
  border: 1px solid var(--border);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.kuz-topbar .t1{
  font-size: 30px;
  letter-spacing: .20em;
  text-transform: uppercase;
  font-weight: 900;
  text-align:center;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

.kuz-topbar .t2{
```

```
  margin-top: 6px;
  font-size: 12px;
  letter-spacing: .26em;
  text-transform: uppercase;
  text-align:center;
  color: var(--muted);
}

.kuz-subbar{
  margin: 0 0 16px;
  padding: 14px 16px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.66), rgba(10,22,44,.52));
  border: 1px solid var(--borderSoft);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.pills{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
}

.pill{
  display:inline-flex;
  align-items:center;
  gap: 8px;
  padding: 9px 12px;
  border-radius: 999px;
  border: 1px solid rgba(0,217,255,.40);
  background: rgba(0,217,255,.08);
  box-shadow: 0 0 10px rgba(0,217,255,.12);
  font-weight: 800;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  color: var(--text);
}
```

```
.pill .k{ color: var(--muted); opacity:.95; font-weight:800; }
.pill .v{ color: var(--text); font-weight:900; text-shadow: var(--glow); }

form.kuz-filters{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
  width:100%;
  margin-top: 12px;
}

/* ========================
   Controls
   ======================== */

select, input[type="text"]{
  background: rgba(4, 10, 20, .55);
  color: var(--text);
  border: 1px solid var(--borderSoft);
  border-radius: 12px;
  padding: 10px 12px;
  outline: none;
}

select:focus, input:focus{
  border-color: var(--border);
  box-shadow: var(--glow);
}

.btn{
  background: rgba(0,217,255,.10);
  color: var(--text);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 10px 14px;
  font-weight: 900;
  letter-spacing: .10em;
  text-transform: uppercase;
```

```
  cursor: pointer;
  transition: transform .12s ease, box-shadow .12s ease, background .12s ease;
}

.btn:hover{
  background: rgba(0,217,255,.18);
  box-shadow: var(--glowStrong);
  transform: translateY(-1px);
}

.btn:active{ transform: translateY(0); }

/* ========================
   KPI
   ======================== */

.kpi{
  min-width: 240px;
  flex: 0 1 auto;
}

.kpi .k{
  color: var(--muted);
  font-weight:800;
  letter-spacing:.04em;
}

.kpi .v{
  margin-top: 6px;
  font-size: 34px;
  font-weight: 900;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

/* ========================
   Tables
   ======================== */

.table-wrap{ overflow:auto; }
```

```
table{
  width:100%;
  border-collapse: collapse;
  border-radius: 14px;
  overflow:hidden;
}

thead th{
  text-align:left;
  padding: 12px 10px;
  color: var(--neon2);
  font-weight: 900;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  background: rgba(0,217,255,.08);
  border-bottom: 1px solid var(--borderSoft);
}

tbody td{
  padding: 12px 10px;
  border-bottom: 1px solid rgba(0,217,255,.14);
  vertical-align: top;
}

tbody tr:hover td{
  background: rgba(0,217,255,.05);
}

code{
  background: rgba(4, 10, 20, .60);
  border: 1px solid rgba(0,217,255,.22);
  border-radius: 10px;
  padding: 4px 8px;
  color: var(--text);
}

/* grids */
.grid-2{
```

```
  display:grid;
  grid-template-columns: 1fr 1fr;
  gap: var(--gap);
}

.grid-3{
  display:grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: var(--gap);
}

@media (max-width: 1100px){
  .grid-3{ grid-template-columns: 1fr; }
}

@media (max-width: 980px){
  .grid-2{ grid-template-columns: 1fr; }
  .kpi{ min-width: 180px; }
}

.footer{
  margin-top: 16px;
  opacity:.86;
}
CSS

# VALIDATE SYNTAX
php -l /var/www/metriks/public/i.php
php -l /var/www/metriks/public/dashboard.php

# RELOAD APACHE
apache2ctl configtest
systemctl reload apache2
```

8.1) /VAR/WWW/METRIKS/PUBLIC/I.PHP

Below are the **current files** (collector + dashboard + CSS) to deploy.

8.1) /VAR/WWW/METRIKS/PUBLIC/I.PHP

```php
<?php
declare(strict_types=1);

/*
  i.php — image endpoint (tracking)
  - Validates HMAC token (sid\npath\nts)
  - Updates daily_site / daily_page / unique_seen
  - Referrer tracking: DISABLED
  - Outputs a 1x1 GIF
*/

$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function outGif(): void {
    header('X-Metriks-Handler: php');
    header('Cache-Control: no-store, no-cache, must-revalidate, max-age=0');
    header('Pragma: no-cache');
    header('Content-Type: image/gif');

    // 1x1 transparent gif
    echo base64_decode('R0lGODlhAQABAPAAAAAAAAAAACH5BAEAAAAALAAAAAABAAEAAAICRAEAOw==');
    exit;
}

function getClientIp(): ?string {
    // Prefer first IP from X-Forwarded-For if present (proxy)
    if (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
        $parts = explode(',', (string)$_SERVER['HTTP_X_FORWARDED_FOR']);
        $ip = trim((string)($parts[0] ?? ''));
        if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
            return $ip;
        }
    }

    $ip = $_SERVER['REMOTE_ADDR'] ?? '';
    if ($ip !== '' && filter_var($ip, FILTER_VALIDATE_IP)) {
```

```
        return (string)$ip;
    }

    return null;
}

function clampStr(?string $s, int $max): string {
    $s = (string)($s ?? '');
    $s = trim($s);
    if ($s === '') {
        return '';
    }
    if (mb_strlen($s, 'UTF-8') > $max) {
        $s = mb_substr($s, 0, $max, 'UTF-8');
    }
    return $s;
}

try {
    $sid = isset($_GET['sid']) ? (int)$_GET['sid'] : 0;
    $p   = isset($_GET['p']) ? (string)$_GET['p'] : '';
    $ts  = isset($_GET['ts']) ? (string)$_GET['ts'] : '';
    $t   = isset($_GET['t']) ? (string)$_GET['t'] : '';

    if ($sid <= 0 || $p === '' || $ts === '' || $t === '') {
        outGif();
    }

    // Normalize path
    if ($p[0] !== '/') {
        $p = '/' . $p;
    }
    $p = preg_replace('#[\x00-\x1F\x7F]#u', '', $p) ?? $p;
    $p = clampStr($p, 1024);

    // ts sanity
    if (!ctype_digit($ts)) {
        outGif();
    }
    $tsInt = (int)$ts;
```

```
$now = time();

// Anti-replay window (±2h)
if ($tsInt < ($now - 7200) || $tsInt > ($now + 7200)) {
    outGif();
}

// Load site secret
$stmt = $pdo->prepare('SELECT secret FROM sites WHERE id = :sid LIMIT 1');
$stmt->execute([':sid' => $sid]);
$secret = $stmt->fetchColumn();
if ($secret === false) {
    outGif();
}
$secretBin = (string)$secret; // BINARY(32)

// Verify HMAC (sid\npath\nts)
$msg = $sid . "\n" . $p . "\n" . $tsInt;
$exp = hash_hmac('sha256', $msg, $secretBin, false);
if (!hash_equals($exp, $t)) {
    outGif();
}

// Visitor identity (unique per day + page)
$ipStr = getClientIp();
$ua    = clampStr($_SERVER['HTTP_USER_AGENT'] ?? '', 512);

$vhInput = ($ipStr ?? '') . "\n" . $ua;
$visitorHash = hash('sha256', $vhInput, true); // BINARY(32)

// daily_site: hits always
$pdo->exec(
    'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
    . 'VALUES (' . (int)$sid . ', CURDATE(), 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);

// daily_page: hits always
$stmt = $pdo->prepare(
    'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
```

```
    . 'VALUES (:sid, CURDATE(), :path, 1, 0)\n'
    . 'ON DUPLICATE KEY UPDATE hits = hits + 1'
);
$stmt->execute([':sid' => $sid, ':path' => $p]);

// IP packed (optional)
$ipBin = null;
if ($ipStr !== null) {
    $packed = @inet_pton($ipStr);
    if ($packed !== false) {
        $ipBin = $packed;
    }
}

// unique_seen: one row per (visitor_hash + path) per day
$scope = 'page';

$stmt = $pdo->prepare(
    'INSERT INTO unique_seen\n'
    . '  (site_id, day, scope, path, visitor_hash, last_seen_at, ip, ua, hits)\n'
    . 'VALUES\n'
    . '  (:sid, CURDATE(), :scope, :path, :vh, FROM_UNIXTIME(:ts), :ip, :ua, 1)\n'
    . 'ON DUPLICATE KEY UPDATE\n'
    . '  last_seen_at = VALUES(last_seen_at),\n'
    . '  ip = VALUES(ip),\n'
    . '  ua = VALUES(ua),\n'
    . '  hits = hits + 1'
);
$stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
$stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
$stmt->bindValue(':path', $p, PDO::PARAM_STR);
$stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
$stmt->bindValue(':ts', $tsInt, PDO::PARAM_INT);
$stmt->bindValue(':ip', $ipBin, $ipBin === null ? PDO::PARAM_NULL : PDO::PARAM_LOB);
$stmt->bindValue(':ua', $ua, PDO::PARAM_STR);
$stmt->execute();

// Detect if it was a new unique for that page today (hits == 1)
$stmt = $pdo->prepare(
    'SELECT hits FROM unique_seen\n'
```

```
        . 'WHERE site_id=:sid AND day=CURDATE() AND scope=:scope AND visitor_hash=:vh AND
path=:path\n'
        . 'LIMIT 1'
    );
    $stmt->bindValue(':sid', $sid, PDO::PARAM_INT);
    $stmt->bindValue(':scope', $scope, PDO::PARAM_STR);
    $stmt->bindValue(':vh', $visitorHash, PDO::PARAM_LOB);
    $stmt->bindValue(':path', $p, PDO::PARAM_STR);
    $stmt->execute();
    $h = (int)$stmt->fetchColumn();

    if ($h === 1) {
        // daily_site uniques +1
        $pdo->exec(
            'INSERT INTO daily_site (site_id, day, hits, uniques)\n'
            . 'VALUES (' . (int)$sid . ', CURDATE(), 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );

        // daily_page uniques +1
        $stmt = $pdo->prepare(
            'INSERT INTO daily_page (site_id, day, path, hits, uniques)\n'
            . 'VALUES (:sid, CURDATE(), :path, 0, 1)\n'
            . 'ON DUPLICATE KEY UPDATE uniques = uniques + 1'
        );
        $stmt->execute([':sid' => $sid, ':path' => $p]);
    }

    outGif();

} catch (Throwable $e) {
    error_log('[metriks i.php] ' . $e->getMessage());
    outGif();
}
```

8.2) /VAR/WWW/METRIKS/PUBLIC/DASHBOARD.PHP

```
<?php
declare(strict_types=1);
```

```
$config = require __DIR__ . '/../config/config.php';
require __DIR__ . '/../src/Db.php';

$db  = new Db($config['db']);
$pdo = $db->pdo();

function h(string $s): string { return htmlspecialchars($s, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-
8'); }
function fmtIp(?string $bin): string {
    if ($bin === null || $bin === '') return '';
    $ip = @inet_ntop($bin);
    return $ip === false ? '' : $ip;
}

$sites = $pdo->query('SELECT id, host, label FROM sites ORDER BY id')-
>fetchAll(PDO::FETCH_ASSOC);

$siteId = isset($_GET['site']) ? (int)$_GET['site'] : (int)($sites[0]['id'] ?? 1);
$range  = (string)($_GET['range'] ?? '7d'); // today|7d|30d
if (!in_array($range, ['today', '7d', '30d'], true)) {
    $range = '7d';
}

$fromExpr = match ($range) {
    'today' => 'CURDATE()',
    '7d'    => 'DATE_SUB(CURDATE(), INTERVAL 6 DAY)',
    '30d'   => 'DATE_SUB(CURDATE(), INTERVAL 29 DAY)',
};

$siteLabel = 'site';
foreach ($sites as $s) {
    if ((int)$s['id'] === $siteId) {
        $siteLabel = (string)($s['label'] ?: $s['host']);
        break;
    }
}

// Daily series
$stmt = $pdo->prepare(
    'SELECT ds.day, ds.hits, ds.uniques\n'
```

```
    . 'FROM daily_site ds\n'
    . 'WHERE ds.site_id = :sid AND ds.day >= ' . $fromExpr . '\n'
    . 'ORDER BY ds.day DESC'
);
$stmt->execute([':sid' => $siteId]);
$series = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Top pages
$stmt = $pdo->prepare(
    'SELECT dp.path, SUM(dp.hits) AS hits, SUM(dp.uniques) AS uniques\n'
    . 'FROM daily_page dp\n'
    . 'WHERE dp.site_id = :sid AND dp.day >= ' . $fromExpr . '\n'
    . 'GROUP BY dp.path\n'
    . 'ORDER BY hits DESC\n'
    . 'LIMIT 50'
);
$stmt->execute([':sid' => $siteId]);
$topPages = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Latest visits (based on unique_seen.last_seen_at)
$latestVisits = [];
try {
    $stmt = $pdo->prepare(
        'SELECT path, last_seen_at, ip, ua, hits\n'
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'ORDER BY last_seen_at DESC\n'
        . 'LIMIT 50'
    );
    $stmt->execute([':sid' => $siteId]);
    $latestVisits = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $latestVisits = [];
}

// User agents (top UA)
$userAgents = [];
try {
    $stmt = $pdo->prepare(
        'SELECT ua, SUM(hits) AS hits\n'
```

```
        . 'FROM unique_seen\n'
        . 'WHERE site_id = :sid AND day >= ' . $fromExpr . '\n'
        . 'GROUP BY ua\n'
        . 'ORDER BY hits DESC\n'
        . 'LIMIT 20'
    );
    $stmt->execute([':sid' => $siteId]);
    $userAgents = $stmt->fetchAll(PDO::FETCH_ASSOC);
} catch (Throwable) {
    $userAgents = [];
}

$sumHits = 0;
$sumUniq = 0;
foreach ($series as $r) {
    $sumHits += (int)$r['hits'];
    $sumUniq += (int)$r['uniques'];
}

// Endpoint detection (for footer)
$endpoint = '/i.gif';
$handler = '';
try {
    $ch = curl_init('https://' . ($_SERVER['HTTP_HOST'] ?? 'localhost') . '/i.gif?x=' .
time());
    curl_setopt_array($ch, [
        CURLOPT_NOBODY => true,
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_TIMEOUT => 2,
        CURLOPT_SSL_VERIFYPEER => false,
        CURLOPT_SSL_VERIFYHOST => false,
    ]);
    curl_exec($ch);
    $handler = (string)curl_getinfo($ch, CURLINFO_HEADER_OUT);
    curl_close($ch);
} catch (Throwable) {
}
?>
<!doctype html>
<html lang="en">
```

```
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Metriks — <?=h($siteLabel)?></title>
  <link rel="stylesheet" href="/assets/css/kuzai-theme.css">
</head>
<body>

  <div class="kz-wrap">

    <header class="kz-header card">
      <div class="kz-header-title">METRIKS</div>
      <div class="kz-header-sub">Dashboard — <?=h($siteLabel)?></div>
    </header>

    <section class="card kz-toolbar">
      <form class="kz-toolbar-form" method="get" action="/dashboard.php">
        <label class="muted">Site</label>
        <select name="site">
          <?php foreach ($sites as $s): ?>
            <option value="<?= (int)$s['id'] ?>" <?= ((int)$s['id'] === $siteId ? 'selected' :
'') ?>>
              <?=h((string)($s['label'] ?: $s['host']))?>
            </option>
          <?php endforeach; ?>
        </select>

        <label class="muted">Range</label>
        <select name="range">
          <option value="today" <?=($range===='today'?'selected':'')?>>Today</option>
          <option value="7d" <?=($range===='7d'?'selected':'')?>>7 days</option>
          <option value="30d" <?=($range===='30d'?'selected':'')?>>30 days</option>
        </select>

        <button class="btn" type="submit">Show</button>
      </form>
    </section>

    <section class="kz-kpis">
      <div class="card kpi">
```

```
      <div class="muted">Hits (range)</div>
      <div class="kpi-value"><?= (int)$sumHits ?></div>
    </div>
    <div class="card kpi">
      <div class="muted">Uniques (range)</div>
      <div class="kpi-value"><?= (int)$sumUniq ?></div>
    </div>
    <div class="card kpi">
      <div class="muted">Days</div>
      <div class="kpi-value"><?= count($series) ?></div>
    </div>
  </section>

  <section class="kz-grid">
    <div class="card">
      <div class="card-title">Daily series</div>
      <table>
        <thead>
          <tr><th>Day</th><th>Hits</th><th>Uniques</th></tr>
        </thead>
        <tbody>
          <?php foreach ($series as $r): ?>
            <tr>
              <td><?=h((string)$r['day'])?></td>
              <td><?= (int)$r['hits'] ?></td>
              <td><?= (int)$r['uniques'] ?></td>
            </tr>
          <?php endforeach; ?>
          <?php if (!$series): ?>
            <tr><td colspan="3" class="muted">No data</td></tr>
          <?php endif; ?>
        </tbody>
      </table>
    </div>

    <div class="card">
      <div class="card-title">Top pages</div>
      <table>
        <thead>
          <tr><th>Path</th><th>Hits</th><th>Uniques</th></tr>
```

```
    </thead>
    <tbody>
      <?php foreach ($topPages as $r): ?>
        <tr>
          <td><?=h((string)$r['path'])?></td>
          <td><?= (int)$r['hits'] ?></td>
          <td><?= (int)$r['uniques'] ?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$topPages): ?>
        <tr><td colspan="3" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
  <div class="card-title">Latest visits</div>
  <table>
    <thead>
      <tr><th>Time</th><th>Path</th><th>IP</th><th>Hits</th><th>User-Agent</th></tr>
    </thead>
    <tbody>
      <?php foreach ($latestVisits as $r): ?>
        <tr>
          <td><?=h((string)$r['last_seen_at'])?></td>
          <td><?=h((string)$r['path'])?></td>
          <td><?=h(fmtIp($r['ip'] ?? null))?></td>
          <td><?= (int)($r['hits'] ?? 0) ?></td>
          <td class="ua"><?=h((string)($r['ua'] ?? ''))?></td>
        </tr>
      <?php endforeach; ?>
      <?php if (!$latestVisits): ?>
        <tr><td colspan="5" class="muted">No data</td></tr>
      <?php endif; ?>
    </tbody>
  </table>
</div>

<div class="card">
```

```
        <div class="card-title">User agents</div>
        <table>
          <thead>
            <tr><th>User-Agent</th><th>Hits</th></tr>
          </thead>
          <tbody>
            <?php foreach ($userAgents as $r): ?>
              <tr>
                <td class="ua"><?=h((string)$r['ua'])?></td>
                <td><?= (int)$r['hits'] ?></td>
              </tr>
            <?php endforeach; ?>
            <?php if (!$userAgents): ?>
              <tr><td colspan="2" class="muted">No data</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>

    </section>

    <footer class="kz-footer muted">
      Endpoint: <code>/i.gif</code> → <code>/i.php</code> • DB: <code>metrics</code>
    </footer>

  </div>

</body>
</html>
```

8.3) /VAR/WWW/METRIKS/PUBLIC/ASSETS/CSS/KUZAI-THEME.CSS

```
/* ========================
   KUZAI / KUZFOLLOW THEME
   neon white + blue
   ======================== */

:root{
  --bg0:#050914;
  --panel: rgba(10, 22, 44, 0.68);
```

```
  --panel2: rgba(10, 22, 44, 0.55);

  --text:#E8F6FF;
  --muted:#B9D7EA;

  --neon:#00D9FF;
  --neon2:#7BE9FF;
  --border: rgba(0, 217, 255, 0.55);
  --borderSoft: rgba(0, 217, 255, 0.28);

  --glow: 0 0 10px rgba(0,217,255,.45),
          0 0 22px rgba(0,217,255,.22);
  --glowStrong: 0 0 14px rgba(0,217,255,.60),
                0 0 34px rgba(0,217,255,.28);

  --radius: 16px;
  --radius2: 22px;

  --pad: 18px;
  --gap: 16px;
}

*{ box-sizing:border-box; }
html,body{ height:100%; }

body.kuzai-metriks{
  margin:0;
  color:var(--text);
  font-family: Inter, system-ui, -apple-system, Segoe UI, Roboto, Arial, sans-serif;

  background-color: var(--bg0);
  background-image: url("/assets/img/FOND-GITHUB-SEARCH2.png");
  background-size: cover;
  background-position: center;
  background-attachment: fixed;
  background-repeat:no-repeat;

  min-height:100vh;
}
```

```
body.kuzai-metriks::before{
  content:"";
  position:fixed;
  inset:0;
  background:
    radial-gradient(1200px 600px at 12% 12%, rgba(0,217,255,.18), transparent 55%),
    radial-gradient(1000px 700px at 85% 25%, rgba(123,233,255,.12), transparent 60%),
    linear-gradient(180deg, rgba(3,6,14,.58), rgba(3,6,14,.78));
  z-index:-2;
}

body.kuzai-metriks::after{
  content:"";
  position:fixed;
  inset:-10%;
  background:
    radial-gradient(900px 420px at 50% 10%, rgba(0,217,255,.10), transparent 60%),
    radial-gradient(1100px 650px at 50% 95%, rgba(0,217,255,.06), transparent 68%);
  filter: blur(2px);
  z-index:-3;
}

a{ color: var(--neon); text-decoration:none; }
a:hover{ text-shadow: var(--glow); }

.kuz-container{
  max-width: 1320px;
  margin: 0 auto;
  padding: 18px 18px 28px;
}

.row{
  display:flex;
  gap: var(--gap);
  flex-wrap: wrap;
  align-items: center;
}

.card{
  background: var(--panel);
```

```
  border: 1px solid var(--borderSoft);
  border-radius: var(--radius);
  padding: var(--pad);
  box-shadow: var(--glow);
  backdrop-filter: blur(8px);
}

.card--thin{ padding: 12px 14px; }

.title{
  font-weight:900;
  letter-spacing:.08em;
  text-transform: uppercase;
  color: var(--text);
  text-shadow: var(--glow);
}

.subtitle, .muted{
  color: var(--muted);
  opacity: .92;
}

/* ========================
   KUZFOLLOW-like top bars
   ======================== */

.kuz-topbar{
  margin: 18px 0 14px;
  padding: 16px 20px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.72), rgba(10,22,44,.58));
  border: 1px solid var(--border);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.kuz-topbar .t1{
  font-size: 30px;
  letter-spacing: .20em;
  text-transform: uppercase;
```

```
  font-weight: 900;
  text-align:center;
  color: var(--text);
  text-shadow: var(--glowStrong);
}

.kuz-topbar .t2{
  margin-top: 6px;
  font-size: 12px;
  letter-spacing: .26em;
  text-transform: uppercase;
  text-align:center;
  color: var(--muted);
}

.kuz-subbar{
  margin: 0 0 16px;
  padding: 14px 16px;
  border-radius: var(--radius2);
  background: linear-gradient(180deg, rgba(10,22,44,.66), rgba(10,22,44,.52));
  border: 1px solid var(--borderSoft);
  box-shadow: var(--glow);
  backdrop-filter: blur(10px);
}

.pills{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
}

.pill{
  display:inline-flex;
  align-items:center;
  gap: 8px;
  padding: 9px 12px;
  border-radius: 999px;
  border: 1px solid rgba(0,217,255,.40);
  background: rgba(0,217,255,.08);
```

```
  box-shadow: 0 0 10px rgba(0,217,255,.12);
  font-weight: 800;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  color: var(--text);
}

.pill .k{ color: var(--muted); opacity:.95; font-weight:800; }
.pill .v{ color: var(--text); font-weight:900; text-shadow: var(--glow); }

form.kuz-filters{
  display:flex;
  flex-wrap:wrap;
  gap: 10px;
  align-items:center;
  width:100%;
  margin-top: 12px;
}

/* ========================
   Controls
   ======================== */

select, input[type="text"]{
  background: rgba(4, 10, 20, .55);
  color: var(--text);
  border: 1px solid var(--borderSoft);
  border-radius: 12px;
  padding: 10px 12px;
  outline: none;
}

select:focus, input:focus{
  border-color: var(--border);
  box-shadow: var(--glow);
}

.btn{
  background: rgba(0,217,255,.10);
```

```
  color: var(--text);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 10px 14px;
  font-weight: 900;
  letter-spacing: .10em;
  text-transform: uppercase;
  cursor: pointer;
  transition: transform .12s ease, box-shadow .12s ease, background .12s ease;
}

.btn:hover{
  background: rgba(0,217,255,.18);
  box-shadow: var(--glowStrong);
  transform: translateY(-1px);
}

.btn:active{ transform: translateY(0); }

/* ========================
   KPI
   ======================== */

.kpi{
  min-width: 240px;
  flex: 0 1 auto;
}

.kpi .k{
  color: var(--muted);
  font-weight:800;
  letter-spacing:.04em;
}

.kpi .v{
  margin-top: 6px;
  font-size: 34px;
  font-weight: 900;
  color: var(--text);
  text-shadow: var(--glowStrong);
```

```
}

/* ========================
   Tables
   ======================== */

.table-wrap{ overflow:auto; }

table{
  width:100%;
  border-collapse: collapse;
  border-radius: 14px;
  overflow:hidden;
}

thead th{
  text-align:left;
  padding: 12px 10px;
  color: var(--neon2);
  font-weight: 900;
  letter-spacing: .08em;
  text-transform: uppercase;
  font-size: 12px;
  background: rgba(0,217,255,.08);
  border-bottom: 1px solid var(--borderSoft);
}

tbody td{
  padding: 12px 10px;
  border-bottom: 1px solid rgba(0,217,255,.14);
  vertical-align: top;
}

tbody tr:hover td{
  background: rgba(0,217,255,.05);
}

code{
  background: rgba(4, 10, 20, .60);
  border: 1px solid rgba(0,217,255,.22);
```

```
  border-radius: 10px;
  padding: 4px 8px;
  color: var(--text);
}

/* grids */
.grid-2{
  display:grid;
  grid-template-columns: 1fr 1fr;
  gap: var(--gap);
}

.grid-3{
  display:grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: var(--gap);
}

@media (max-width: 1100px){
  .grid-3{ grid-template-columns: 1fr; }
}

@media (max-width: 980px){
  .grid-2{ grid-template-columns: 1fr; }
  .kpi{ min-width: 180px; }
}

.footer{
  margin-top: 16px;
  opacity:.86;
}
```

9) POST-DEPLOY VERIFICATION CHECKLIST

```
# APACHE CONFIG SANITY
apache2ctl configtest
systemctl reload apache2

# COLLECTOR ENDPOINT (MUST BE PHP)
curl -I "https://metriks.kuzlab.org/i.gif?x=$(date +%s)" | head -n 40
```

```
# DASHBOARD
curl -I "https://metriks.kuzlab.org/dashboard.php" | head -n 25

# DB GROWTH
mysql -N metrics -e "SELECT COUNT(*) AS daily_site_rows FROM daily_site;"
mysql -N metrics -e "SELECT COUNT(*) AS daily_page_rows FROM daily_page;"
mysql -N metrics -e "SELECT COUNT(*) AS unique_seen_rows FROM unique_seen;"
```